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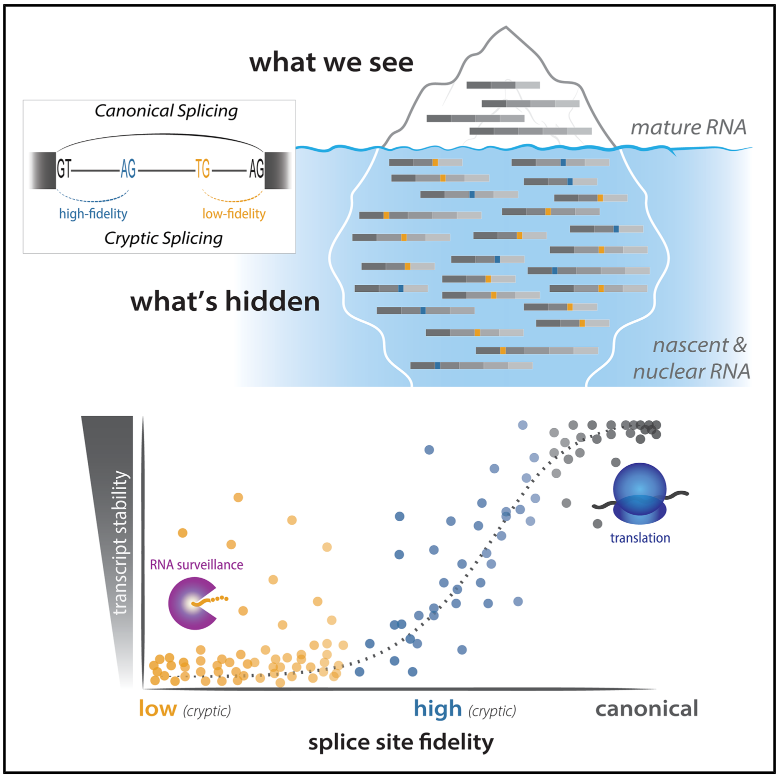
Pervasive noise in human pre-mRNA splice site selection

Graphical abstract



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In brief

Human splicing is noisy, with cryptic splicing at both high- and low-fidelity motifs, associated with features that probabilistically increase stochastic splice site usage. These findings reveal a distinct layer of noise in the regulation of gene expression.

Highlights

- Nascent RNA sequencing reveals pervasive cryptic splicing in human transcriptomes
- Gene length and transcriptional activity are predictive of cryptic splicing levels
- Low-fidelity splice site motifs are frequently used and preferentially degraded
- RNA surveillance pathways shape the output of splicing noise

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Pervasive noise in human pre-mRNA splice site selection

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SUMMARY

RNA splicing has historically been thought to be highly efficient and accurate, with little opportunity for deviation from regulated alternative splicing. This dogma has been challenged by recent observations that biological noise may contribute substantially to transcriptome diversity. However, quantitative understanding of stochastic splicing variation is challenging because these transcripts are likely subject to rapid degradation. Here, we use deep sequencing across RNA compartments to track splicing intermediates in human cells and see abundant cryptic splicing associated with genomic features that promote splicing noise. We observe pervasive usage of low-fidelity splice sites, likely due to stochasticity in recruitment or binding of the spliceosome. These sites are turned over quickly and show evidence for nuclear and cytoplasmic degradation, suggesting widespread surveillance and rapid quality control of non-productive transcripts. Our findings provide insights into the propensity for error in RNA processing mechanisms and regulation of alternative splice sites across a gene.

INTRODUCTION

Pre-messenger RNA (pre-mRNA) splicing is a crucial step in metazoan gene regulation, with 95% of genes containing introns that are excised to produce mature transcripts. Intron excision is performed by the spliceosome, a highly conserved molecular machine assembled from numerous small nuclear RNAs (snRNAs) and proteins. Spliceosome recruitment to pre-mRNA molecules is initiated by binding of U1 snRNA to the 5′ splicing donor site (GT or GC), U2 snRNA to the branchpoint A, and U2AF1 to the 3′ splicing acceptor site (AG). Less than 1% of mammalian introns are spliced using the minor spliceosome, involving U11 snRNA binding to an AT 5′ splicing donor, U12 snRNA to the branchpoint A, and ZRSR2 to the AC 3′ splicing acceptor site. Both major and minor spliceosome-regulated splice sites are among the most conserved metazoan genetic elements,[1–3] leading to a prevalent perspective that the spliceosome is a high-fidelity, deterministic enzyme.[4,5] However, core dinucleotides occur within longer, more degenerate consensus motifs[6] for a total of ∼30 informative nucleotides that are often thousands of nucleotides apart, and alternative splice sites are regulated by auxiliary sequence elements that recruit splicing regulatory factors (SRFs).[7,8] Finally, most splicing is thought to occur co-transcriptionally, with splice site recognition occurring relatively rapidly as RNA polymerase II (RNAPII) is still transcribing.[9,10] The accuracy of splicing decisions relies on the combined specificity of these genetic and biochemical interactions to assemble the spliceosome and catalyze intron excision, providing ample opportunities for stochastic interactions that might result in undesirable mRNA products.

Studies in the last few decades have started to investigate the mechanisms of erroneous splicing and quality control to purge incorrectly spliced transcripts. Unlike DNA replication and transcription or translation mechanisms, which have more narrowly defined error rates (10^{-9} base pairs and 10^{-6}–10^{-5} nucleotides or amino acids, respectively[11–14]), less is known about splicing error rates (estimated to be between 10^{-5} and 10^{-2} junctions).[15–17] Early steps to minimize splicing error include kinetic proofreading, in which DExD/H-box ATPases outcompete suboptimal substrates and promote spliceosomal conformations for high-fidelity splice site usage.[16,18,19] The spliceosome is thought to iteratively sample potential splice sites and uses kinetic proofreading to reject suboptimal sites before settling on an optimal site, similar to promoter proofreading by RNAPII.[20,21] This process is not perfect, with many nuclear or cytoplasmic "spell-checking" mechanisms to recognize and clear aberrantly or unspliced RNAs that escape earlier proofreading checkpoints.[19] Nuclear turnover involves exosome or

XRN2 degradation, both of which compete with splicing on pre-mRNA.[18,22] In the cytoplasm, turnover is often translation-dependent, with either nonsense-mediated decay (NMD) or non-stop decay (NSD) triggered after the first round of ribosome translation.[23,24]

Despite extensive quality control steps to mitigate erroneous splice site usage, there is increasing evidence that "noisy" splicing regularly occurs at cryptic or low-fidelity sites. Previous studies have defined splicing noise as (1) minor alternatively spliced isoforms from canonical splice sites,[2,25] (2) unannotated splice sites used at very low frequency and/or rapidly degraded,[15,26] and (3) so-called low-fidelity splicing at non-consensus motifs.[17,26] Additionally, non-canonical splice site usage can occur through recursive splicing, which is the use of juxtaposed 5′/3′ splice sites to trigger piecewise excision of long introns.[27,28] Recursive splicing might be the result of stochastic interactions that increase splicing efficiency, especially in human cells,[29] perhaps by reducing noisy splice site usage.[30] The lack of a consistent definition for what constitutes noise in splicing has made it difficult to understand the full landscape of potential splicing interactions. Many studies have identified extensive non-canonical splicing of various forms across metazoan cells,[4,26,30–33] but it is challenging to directly compare across studies given the diversity of techniques applied to estimate non-canonical splice site usage. Thus, the intrinsic levels of splicing noise and the ways in which cells are able to reduce the burden of this apparently wasted transcriptional output are unclear.[4]

We set out to answer a fundamental question in RNA biology: to what extent does stochastic splicing occur in mammalian cells? Here, using deep sequencing data to track splicing intermediates across RNA lifecycles, we systematically investigate the usage of cryptic and low-fidelity splice sites to understand noise in mRNA splicing. Our analyses reveal pervasive expression of cryptic sites in nascent RNA populations that does not persist to mature RNA, even after inhibition of translation-dependent degradation processes. We further show that, though most cryptic splicing likely occurs stochastically, we can identify genomic features that predict the occurrence or fate of a subset of these cryptic events.

RESULTS

Splice site selection is thought to occur mostly co-transcriptionally, soon after sites become available on pre-mRNA transcripts.[9] To identify cryptic events independent of transcript localization or stability effects, we selectively enriched for newly transcribed RNA. Specifically, we isolated nascent RNA from the nucleus after rapid metabolic labeling with 4-thio-uridine (4sU) for 8 minutes. We compared nascent RNA with total nuclear RNA, polyA-selected RNA, and polyA-selected RNA after cycloheximide (CHX) treatment to inhibit translation-mediated degradation mechanisms (e.g., NMD) (Figure 1A). We conducted these experiments independently in K562 erythroleukemia and KNS60 glioblastoma cell lines and performed high-throughput sequencing (4 replicates each except for CHX treatment; STAR Methods; Table S1). We validated the high quality of the RNA enrichment methods: (1) western blots of nuclear markers confirmed efficient nuclear fractionation (Figure S1A), (2) nascent and nuclear samples showed high levels of intronic reads from pre-mRNA molecules (Figure S1B), and (3) samples clustered by their RNA enrichment method (Figure S1C).

Splice site positions can be inferred using junction reads split across exon-exon boundaries. However, technical or biological artifacts (e.g., overlapping reads, mapping irregularities, or sequence polymorphisms) may lead to mis-quantification or mis-identification of splicing events (Figure S2A). We developed a computational framework called cryptic identification of splice sites (*CRYPTID-SS*) to process RNA sequencing (RNA-seq) reads, identify high-confidence splice sites, and quantify their usage (STAR Methods). To avoid known artifacts, we only considered uniquely mapped junction reads satisfying a number of criteria, including sites within the same annotated gene and not overlapping polymorphisms or repeats (Figure S2B). We categorized canonical or cryptic sites based on annotation status and sequence content of the core splice site dinucleotides. Canonical sites are defined as annotated sites with high-fidelity splice site motifs recognized by either the major or minor spliceosome, while cryptic sites are either not annotated and/or contain low-fidelity dinucleotides (Figure 1B). To enable mechanistic understanding of cryptic site usage, we further categorized cryptic sites into three categories: low-fidelity or high-fidelity splice site dinucleotides or a juxtaposed 3′ and 5′ splice site motifs around a central AG|GT motif supporting the possibility of a recursive splicing event.

Cryptic splicing is pervasive in nascent RNA

We identified 2,110,022 and 2,179,358 independent splice sites across all K562 and KNS60 samples, respectively, approximately evenly split between 5′ and 3′ splice sites. We observe a 41% increase in cryptic splice sites, on average, in nascent and nuclear RNA relative to mature RNA (Figure 1C). This is complemented by a 206% increase, on average, in the number of cryptic junction reads in nascent and nuclear samples compared with mature RNA (Figure 1D). While canonical sites are supported by more than 100 reads on average, cryptic sites rarely have more than 1 supporting read (Figure 1E), indicating that splicing at any given cryptic site occurs very infrequently. Concordantly, 87% of cryptic sites are detected in only one enrichment method (Figure 1F)—in contrast to canonical sites, most of which are detected in all four enrichment methods—and an average of 84% are present in only one replicate from that enrichment method (Figure S3A). We assessed the sensitivity of our detection pipeline by subsampling reads to 10% to 90% of the total reads collected and saw that new cryptic sites were still detected as read depth increased from 50% to 100% of sequenced nascent reads (Figure 1G), and thus would likely increase further at higher read depths. However, the rate of detection starts to plateau in this range, suggesting an upper limit on genomic positions that can be used as cryptic splice sites.

To rule out technical artifacts that may lead to detection of mis-identification of splice sites, we performed the same analyses with a series of control datasets. First, to examine splice site usage upon 4sU treatment, we used polyA RNA from cells treated with uridine and 4sU for 8 minutes or 24 hours and saw no

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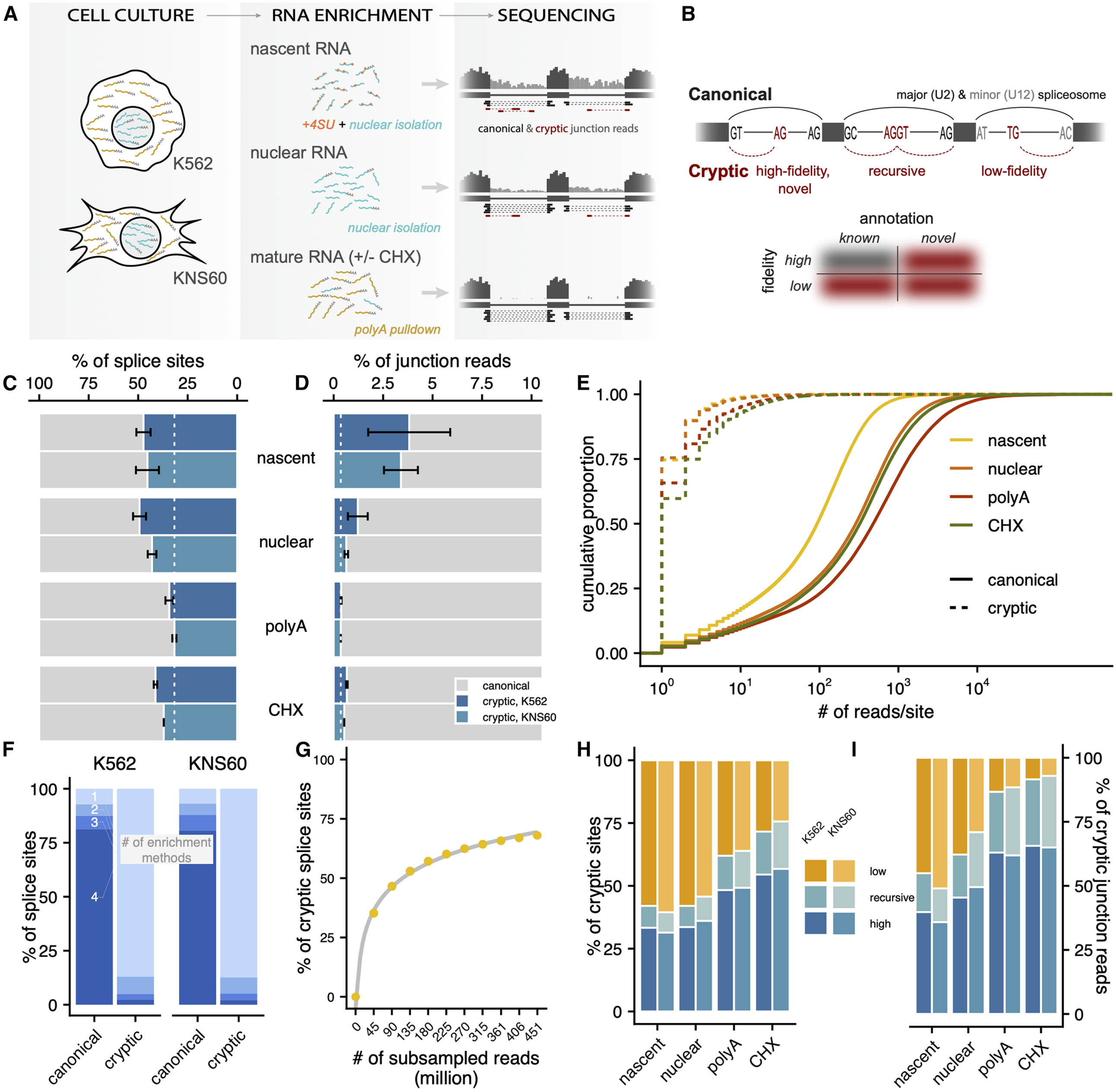

Figure 1. Cryptic splicing is pervasive across enrichment methods

(A) Study design and predicted patterns of high-throughput sequencing reads.

(B) Schematic of canonical (black) and cryptic (red) splicing.

(C and D) Percentage of canonical (gray) and cryptic (blue) splice sites and junction reads detected across enrichment methods and cell types. Dotted white line represents mean percentage for polyA-enriched samples, and error bars represent standard error across replicates.

(E) Cumulative proportion of junction reads per site for canonical (solid) and cryptic (dashed) splice sites.

(F) Percentage of canonical or cryptic splice sites that were detected in 1, 2, 3, or all 4 enrichment methods.

(G) Percentage of cryptic sites (out of all splice sites) detected across various sub-samples of read coverage (in 10% increments) in nascent RNA data.

(H and I) Percentage of cryptic sites and junction reads that can be classified as low-fidelity (orange), recursive (teal), or high-fidelity sites (dark blue).

substantial increase in cryptic sites (Figures S3B and S3C). Second, previous reports have suggested that library preparation may induce spurious junction reads (e.g., through trans-splicing[34]). Using both oligo-dT and random hexamer-primed libraries from synthetic spike-in RNA variant (SIRV) mixes, we see that less than 0.3% of junction reads are erroneous (Figures S3D and S3E). Third, we evaluated mapping accuracy with BLAST-like alignment tool (BLAT), an orthogonal alignment software.

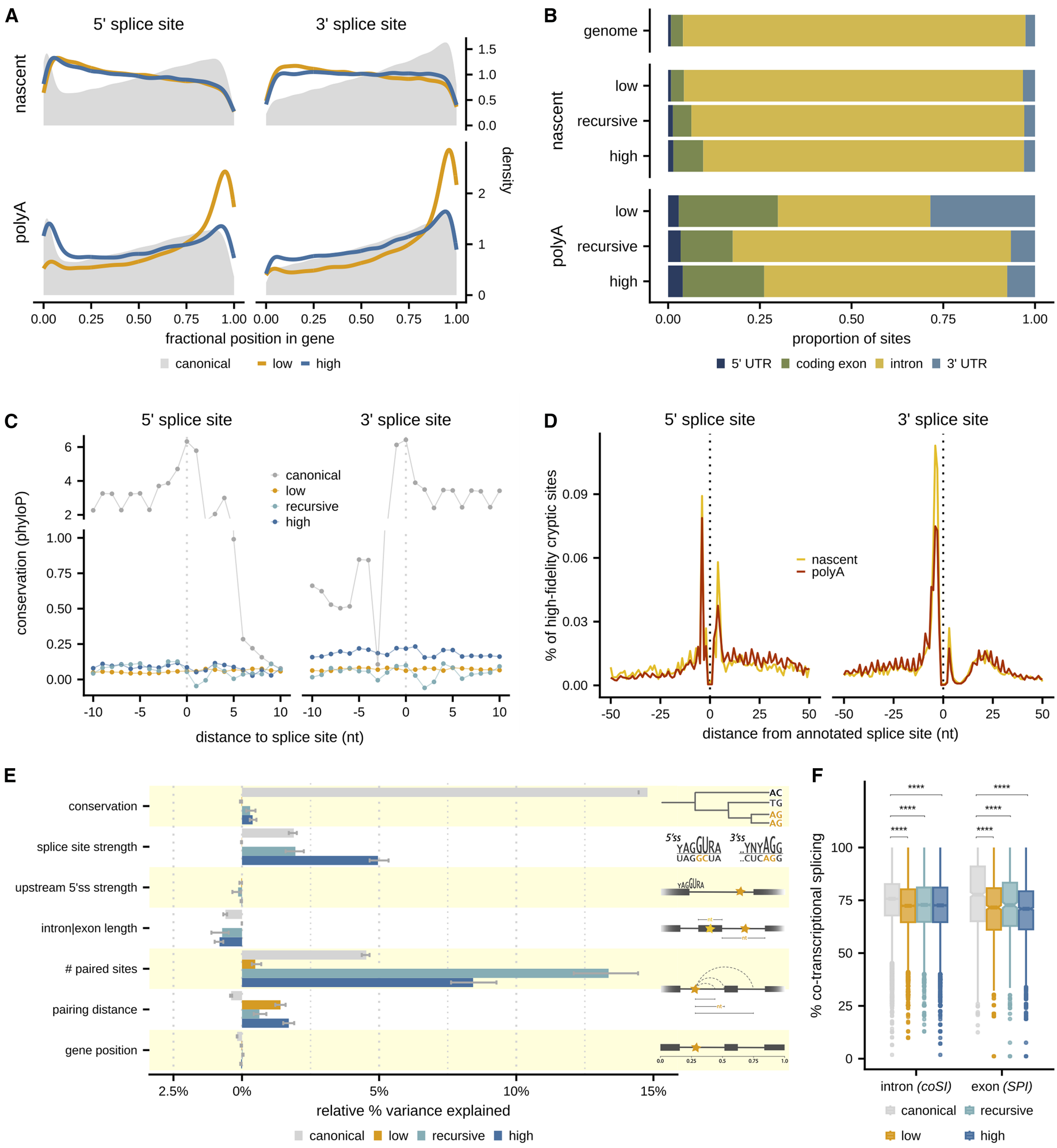

Figure 2. Genomic features influence the use of cryptic sites

(A) Position of splice sites across a gene for canonical (gray), low-fidelity (orange), or high-fidelity (dark blue) sites in either nascent or mature RNA, where fractional position is the distance from the gene's start relative to gene length.

(B) Proportion of cryptic site categories detected within annotated 5′ UTRs (navy blue), coding exons (green), introns (yellow), and 3′ UTRs (light blue) for sites detected in nascent or mature RNA. Top bar shows the proportion of nucleotides within expressed genes for each annotation category.

(C) Nucleotide-specific phyloP scores in a 20 nt window around 5′ and 3′ splice sites for canonical (gray), low-fidelity (orange), recursive (teal), or high-fidelity (dark blue) sites.

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We see that 77.3% of high-fidelity sites and 87.1% of low-fidelity sites are mapped to the same genomic positions with STAR and BLAT, similar to the 86.7% of canonical sites mapped to the same genomic positions. We then evaluated if mapping errors caused cryptic site identification. Using DNA whole-genome sequencing (WGS) from K562 cells where all junction reads are due to sequence variation or mapping biases, we see that, on average, 1.32% of cryptic splice sites in our data are also present in WGS. These sites were removed from the final analyses. We also simulated reads from both pre-mRNA and mRNA transcripts of annotated genes as a ground truth to estimate the fraction of erroneous junctions expected in the sequencing data. Using simulated reads, we estimate that 0.19%–0.35% of cryptic sites, after the filtering steps above, are expected to be false positives based on mapping biases (STAR Methods).

Finally, we chose a subset of cryptic site pairs to validate with orthogonal experiments. Using primers flanking highly expressed junctions using low-fidelity or high-fidelity sites (33 and 35 junctions, respectively) identified in either nuclear or mature RNA (Table S2), we amplified cDNA products from the respective RNA enrichments (STAR Methods). We performed amplicon sequencing on all 33 amplicons with a single band and validated precise positions of cryptic sites for all sites except one of the low-fidelity mature RNA junctions. Using quantitative reverse-transcription PCR (RT-qPCR), we see that the relative expression of cryptic junctions is correlated with the sequencing read counts (Spearman $\rho = 0.71$ and $\rho = 0.37$ for sites identified in nuclear and mature RNA, respectively; Table S3; Figure S3F).

Previous studies hypothesized that most cryptic molecules are degraded by mechanisms like NMD, so blocking translation should increase cryptic splicing in mature RNA.[31,33,35] While we see a higher percentage of cryptic sites and reads in CHX-treated RNA (19% and 61% higher, respectively; Figures 1C and 1D), these increases are modest relative to those seen in nascent RNA. Treatment with a chemical NMD inhibitor (NMDi14) gave consistent results (Figures S3B and S3C). When we sub-categorize cryptic sites, we see that the majority of cryptic sites in nascent RNA are low-fidelity, while the majority of cryptic sites in the CHX-treated RNA are high-fidelity (Figure 1H). Overall, low-fidelity splice site abundance gradually decreases as RNA becomes more mature (Figures 1H and 1I), suggesting these molecules are even less stable than products of other cryptic splicing events. The proportion of sites classified as motif-consistent recursive sites increases as RNA matures, contrary to the expectation that true recursive intermediates would be lost when their parent intron is fully spliced out, and suggests that splicing at motif-consistent recursive sites often results in stable mRNA. Finally, the increase in high-fidelity cryptic splicing in CHX-treated RNA suggests that these sites are the most likely to trigger NMD. Together, these results indicate pervasive cryptic splicing—including at motifs that the spliceosome has not been thought to bind—and that perturbation of translation-dependent degradation mechanisms does not completely uncover the full landscape of splice site usage on pre-mRNA.

Cryptic splicing occurs throughout genes

Our expanded catalog of cryptic splice sites across the RNA lifecycle enables systematic assessment of the regulatory environments in which cryptic splicing occurs. We focused on splice sites that were supported by at least two uniquely mapping reads and/or at least two independent replicates (Table S4). Using these high-confidence sites, we first looked at the distribution of sites within genes. Canonical 5′ss are distributed in a hockey-stick-like distribution (Figure 2A) across a gene, with a 5′ peak followed by a dip and slow increase likely driven by very long first introns in human genes.[36] Correspondingly, canonical 3′ss steadily increase across gene bodies. Cryptic sites in nascent and nuclear RNA are more evenly distributed, with both 5′ and 3′ cryptic sites slightly more likely to occur near the beginning of the gene (Figure 2A; Figure S4A). We see that the distribution of cryptic sites in nascent and nuclear RNA closely matches the distribution of nucleotides across UTRs, coding exons, or intronic regions (Figure 2B; Figure S4B), with a modest enrichment of motif-consistent recursive and high-fidelity sites in coding exons (odds ratio = 1.56–2.56; hypergeometric p value < 2.2×10^{-16} for both cryptic types). In mature RNA, cryptic sites more closely match canonical site distributions and they are significantly depleted in introns (odds ratio = 0.14; hypergeometric p value < 2.2×10^{-16}), with low-fidelity sites sharply enriched at the 3′ end of the gene and in 3′ UTR regions (odd ratio = 14.97; hypergeometric p value < 2.2×10^{-16}; Figures 2A and 2B; Figures S4A and S4B). These observations suggest that while there is promiscuous spliceosomal binding across a gene, the positioning of cryptic sites (particularly in 3′ UTRs) likely influences transcript stability or the potential to trigger NMD.

Canonical splice sites are among the most conserved sites in mammalian genomes. If cryptic sites reflect noisy splicing, rather than regulated splicing, they might not show the same conservation patterns. We see that canonical 5′ and 3′ splice sites are highly conserved (Figure 2C), with sustained conservation into exonic regions exhibiting triplet periodicity due to the 3rd "wobble" nucleotide in a codon.[26] Cryptic sites exhibit little to no conservation signatures, though there is weak evidence for triplet periodicity downstream of 3′ cryptic motif-consistent recursive and high-fidelity sites. While other studies have seen moderate to strong conservation for bona fide recursive sites in mammalian and particularly in fly cells,[30,37–40] recursive sites classified based on motifs alone show no evidence for conservation. The pattern of slightly higher conservation around high-fidelity 3′ss may reflect NAGNAG splicing, a 3′ss mechanism producing two or more alternative isoforms differentiated by three nucleotides (NAG).[41] Indeed, we see that while ∼94% of cryptic splicing occurs more than 50 nucleotides from an annotated

(D) Distribution of high-fidelity cryptic 5′ or 3′ splice sites around the nearest canonical 5′ or 3′ splice site, respectively, in nascent (yellow) or mature RNA (red).
(E) Relative importance of features influencing variance in usage of different splice site categories, using multiple linear regression models. Bars pointing to the right indicate positively correlated features, while bars pointing left indicate negatively correlated features. Error bars represent bootstrapped confidence intervals.
(F) Completed splicing index (coSI) or splicing per intron (SPI) for splice sites in annotated introns or exons, respectively, across splice site categories. Significance assessed with Mann-Whitney U test. **** adjusted p value < 0.0001.

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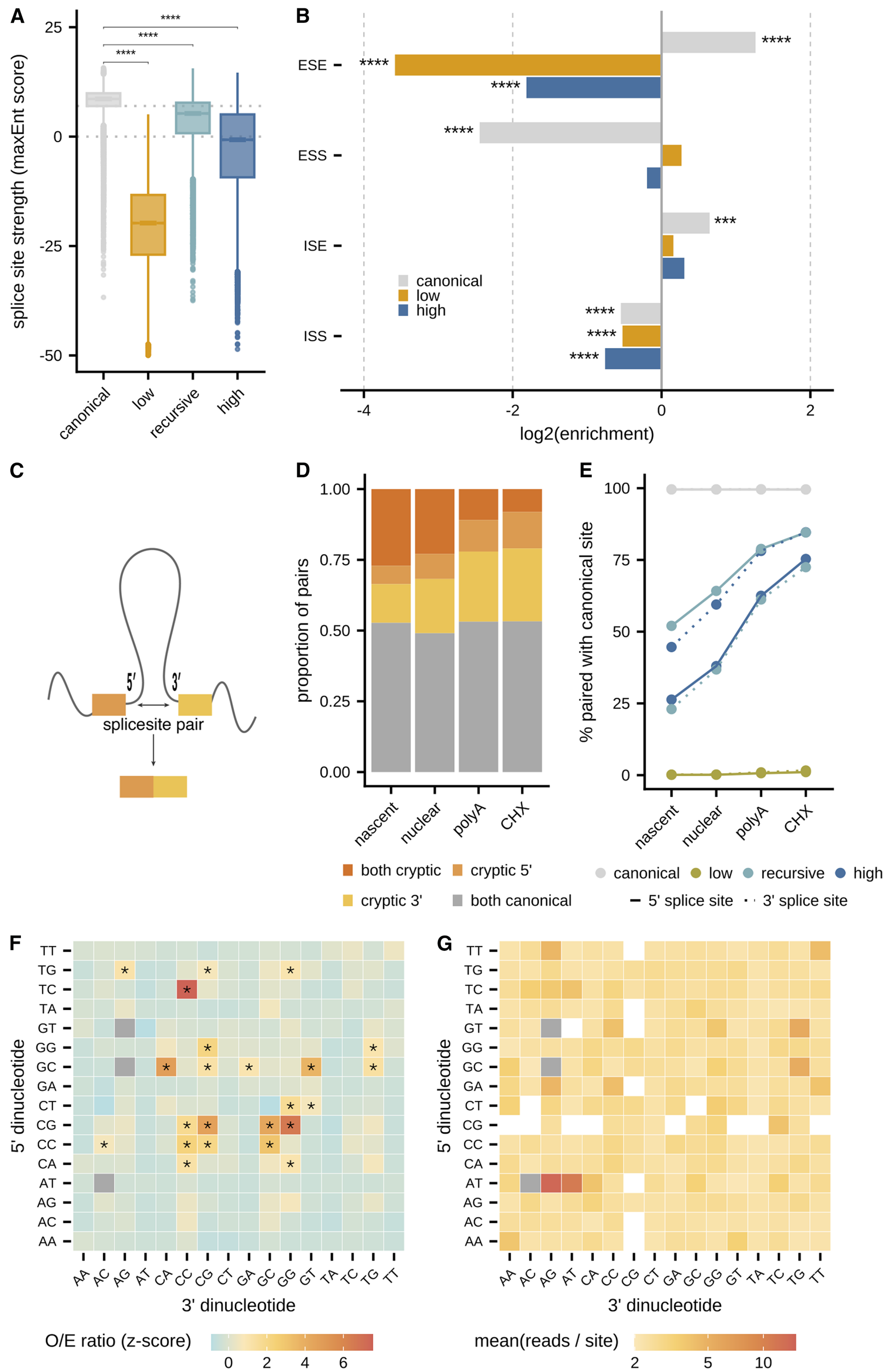

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splice site, high-fidelity cryptic sites are enriched around annotated sites (Figure 2D). This is particularly true for 3′ss, where as many as 10% of high-fidelity 3′ sites might be NAGNAG sites in both nascent and mature RNA. In mature RNA, the frequency of high-fidelity sites around annotated splice sites exhibits a triplet periodicity, suggesting that high-fidelity sites that maintain open reading frames are more likely to result in stable isoforms.

Together, our results thus far suggest that cryptic sites (especially low-fidelity sites) are used at random across a gene. To test this, we used a multiple linear regression framework to estimate how genomic features contribute to the usage of individual cryptic sites (STAR Methods). Fitting this model separately to each cryptic type, we can explain 22% of variance in canonical site usage and 2%–17% of variance in cryptic splicing in nascent RNA (Figure 2E). Canonical site usage is positively associated with conservation and splice site strength. Splice site strength also explains 3%–5% of variance in motif-consistent recursive and high-fidelity site usage (particularly in mature RNA; Figure S4C), consistent with higher spliceosomal binding due to similarity with canonical splice sites. Cryptic sites were also used more if they were paired with more partner sites that were farther away, which was true even when correcting for the number of partner sites (Figure S4D). Low-fidelity sites were less likely to be used when located downstream of strong canonical 5′ss, suggesting that stronger sites might already be committed to a splicing reaction that otherwise might involve a cryptic site. Consistently, cryptic splice sites were more likely to occur in introns or exons with less co-transcriptional splicing (Figure 2F). These observations point to the local splicing environment being an important contributor to cryptic splice site usage, where inefficient canonical splicing might lead to an opening for cryptic sites to pair with a partner site and complete splicing after promiscuous binding of the spliceosome.

Sequence determinants of cryptic splicing

Splice sites are often delineated by core dinucleotide identity, but splice site usage also depends on surrounding sequence context and nearby splicing regulatory elements (SREs). Overall, cryptic splice sites have significantly weaker splice site strengths than canonical sites, and low-fidelity sites are the weakest, as expected given the lack of canonical dinucleotides (Figure 3A; Figure S5A). High-fidelity sites have similar, albeit weaker, nucleotide enrichments as canonical sites across the extended motif region, particularly for the +5 G nucleotide in 5′ss and polypyrimidine tract upstream of 3′ss for sites identified in mature RNA (Figures S5B and S5C). Nascent RNA-identified 3′ss have a weak polypyrimidine tract, mostly composed of cytosines instead of uracils. We computed position weight matrix (PWM) scores[42] (STAR Methods) to quantify the contribution of surrounding sequence to splice site strength. Even without the core dinucleotides, low-fidelity sites have significantly lower PWM scores than canonical or other cryptic sites, suggesting that splicing at these sites is likely not driven by the presence of expected sequences in surrounding nucleotides (Figure S5D). Finally, we see that the putative intronic regions around both low- and high-fidelity cryptic sites are significantly depleted of exonic splicing enhancers (ESEs) and intronic splicing silencers (ISSs) (STAR Methods; Figure 3B), suggesting that cryptic site usage might be due to fewer repressive elements rather than active exon recognition (unlike canonical splice sites).[43]

While 5′ and 3′ splice sites are likely recognized as independent units, their favorable pairing across an intron is necessary for splicing (Figure 3C). The majority of pairs across RNA life cycles involve two canonical sites (Figure 3D). On average, canonical sites pair with sites that are slightly more distant than the closest annotated site, while cryptic sites often pair with sites that are closer than the closest annotated site (Figure S5E). The proportion of cryptic-cryptic site pairing decreases as RNA matures, with 57% cryptic-cryptic pairing in nascent RNA but 83%–87% of cryptic-canonical pairing in mature RNA, indicating that cryptic transcripts have a higher probability of surviving to maturity when cryptic sites are paired with canonical sites. The increase in cryptic-canonical pairing across RNA life cycles mostly reflects pairing propensities of motif-consistent recursive and high-fidelity sites, while almost all low-fidelity sites are paired with other cryptic sites (Figure 3E; Figure S5F). There is more pairing between motif-consistent recursive 3′ and 5′ sites in both nascent and mature RNA (Figure S5F), while motif-consistent recursive 5′ sites are more likely to pair with canonical sites. This indicates that recursive splicing may not occur in a sequential (upstream-segment-first) order across an intron, but rather, consistent with recent reports,[29] individual segments flanked by recursive sites can be spliced out of larger introns.

These observations suggested sequence preferences that may underlie the probability of successful splicing involving cryptic sites. Overall, we do not see enrichments of any particular dinucleotides among low-fidelity sites. Instead, we see that there are 23 pairs of dinucleotides that are significantly enriched, with 2.19-fold more sites on average than expected, relative to their joint occurrence in transcribed regions (Figure 3F; STAR Methods). Among the preferred dinucleotide pairs, 91% are

Figure 3. The pairing of low-fidelity splice sites is driven by specific dinucleotide preferences

(A) Splice site strengths (maxEnt scores) for canonical (gray), low-fidelity (orange), recursive (teal), or high-fidelity (dark blue) splice sites. Significance assessed with Mann-Whitney U test. **** adjusted p value < 0.0001.

(B) Enrichment of different categories of SREs around splice sites from different categories. Significance assessed with hypergeometric test. *** adjusted p value < 0.001 and **** adjusted p value < 0.0001.

(C) Pairing between 5′ and 3′ splice sites defining an excised intron.

(D) Proportion of splice site pairs in which both, only one, or neither of the splice sites are cryptic.

(E) Percentage of 5′ (solid line) or 3′ (dashed line) sites that are paired with canonical sites across splice site categories.

(F) Observed/expected ratios for each combination of 5′ (y axis) and 3′ (x axis) low-fidelity splice site dinucleotides detected in nascent RNA. Canonical dinucleotide pairs were omitted (gray boxes). Significance assessed using permutations. * adjusted p value < 0.05.

(G) Mean number of reads per site for each combination of 5′ (y axis) and 3′ (x axis) low-fidelity splice site dinucleotides detected in nascent RNA. Canonical dinucleotide pairs and pairs with fewer than 25 observations were omitted (gray and white boxes, respectively).

GC-rich, pointing to mRNA secondary structure or stability playing a role in cryptic site usage. G-quadruplex (G4) motifs have been seen to be enriched near intron-exon boundaries, and the formation of G4 structures enhances splice site usage.[44] However, these overrepresented sequence pairs do not represent the most frequently expressed cryptic junctions. Instead, AT-AG and AT-AT pairs are used most often (Figure 3G), suggesting that the minor spliceosome may more frequently use different 3′ss than previously thought.[45] Importantly, we do not see similar enrichments in the SIRV control dataset (Figures S5G and S5H), indicating that these enrichments are not likely due to errors or biases that occur during library preparation, sequencing, or mapping.

Gene length and transcription levels predict cryptic splicing levels across genes

Our results suggest pervasive cryptic splicing driven primarily by stochastic spliceosome binding across pre-mRNA transcripts. If true, cryptic splicing would be more likely to occur in genes whose genomic or transcriptional properties presented more opportunity for splicing noise. For instance, longer genes present greater sequence space for spliceosomal binding at random. Similarly, if every pre-mRNA transcript has the same probability of being cryptic spliced, higher transcription levels would result in more cryptic splicing. However, mRNA transcript composition is highly regulated and under strong evolutionary constraints. Thus, we wondered how variable cryptic splicing was across genes with different genomic features and if genes varied in their tolerance for cryptic splicing.

We first quantified the proportion of cryptic transcripts from each gene with a junction TPM, which calculates the relative expression of each type of cryptic transcript using only exon-exon junction reads (STAR Methods). We ranked genes by these TPMs and saw large variability in the extent of cryptic splicing per gene in nascent RNA. The proportion of cryptic transcripts is negatively correlated with both transcription levels and steady-state gene expression levels (assessed from nascent RNA and mature RNA, respectively; STAR Methods) and the number of introns in a gene but positively correlated with gene length and the fraction of a gene annotated to be intronic (Figure 4A; Table S5). Genes can be clustered into 3 distinct categories: (1) 540 (4.2%) genes with no cryptic splicing, which are lowly transcribed, moderately expressed, short genes enriched for genes involved in oxygen binding and gas transport; (2) 11,071 genes (87%) with cryptic transcripts accounting for 1%–99% of their total transcripts; and (3) 1,053 (8.3%) genes with nearly or exactly 100% cryptic transcripts—including 424 genes with 100% cryptic splicing but no low-fidelity cryptic splicing, which are lowly transcribed, lowly expressed genes with few annotated introns. This last category is enriched for genes involved in olfaction, structural components of chromatin (i.e., histone genes), and electron transport chain activity, suggesting that regulatory pressures on these pathways might be reduced when these genes are lowly expressed.

Still, the presence of genes that lack consensus splicing is surprising. Protein-coding genes have the least cryptic transcripts across all RNA enrichment methods, so there are very few protein-coding genes with no consensus splicing (Figures S6A and S6B). The few protein-coding genes with 100% cryptic transcripts in nascent RNA are mostly spliced in mature RNA, suggesting slower consensus splicing for these genes. While the proportion of protein-coding genes without consensus splicing increases in mature RNA, 40% of those genes are intronless. Intronless genes do have cryptic splicing and tend to use cryptic sites more than genes with one or more introns after normalizing for gene length (Figures S6C and S6D).

While there are clear associations between the extent of cryptic splicing and a gene's genomic features, the high degree of correlations between pairs of these features makes it difficult to disambiguate the relative role of each. To overcome this challenge, we trained an elastic net regression model, which specifically accounts for correlated variables, to predict gene-specific usage of cryptic splice sites based on the combined influence of these genomic features (STAR Methods). We can explain 58%–81% variance in both the number of cryptic sites and the number of cryptic reads for a gene across all types of splice sites for nascent and nuclear RNA (Figure S6E). Transcription levels and gene length contributed the most to cryptic splicing in nascent RNA, both positively (Figure 4B; Figure S6F). Low-fidelity sites are particularly associated with longer gene lengths. Higher mRNA levels were weakly negatively associated only with motif-consistent recursive and high-fidelity splicing, as expected if those transcripts are more likely to be degraded after reaching maturity. By contrast, these features only explain 44%–74% of variance across cryptic sites in mature RNA, with mRNA levels and intron number together explaining 64% of variance in the canonical splice site usage (Figure S6G).

We used our predictive framework to identify genes with more or less cryptic splicing in nascent RNA than expected given their genomic features. Using a conservative threshold to define outliers, we see that more genes (3.8%) have less cryptic splicing than expected and that 81% of these genes are mRNAs (Figure 4C). However, only 1.6% of genes have more cryptic splicing than expected, and these genes are enriched for expressed pseudogenes or long-noncoding RNAs (lncRNAs) (39% of genes; odds ratio = 6.93; hypergeometric p value = $< 2.2 \times 10^{-16}$). This is consistent with these types of RNAs having less splicing regulation and/or less selective constraints on their sequence content,[46,47] leading to increased tolerance for cryptic splicing without severe functional consequences. It has been hypothesized that lncRNA-spliceosome complexes themselves—rather than fully spliced lncRNA—might serve a role in stimulating local gene expression, suggesting that the function of lncRNA splicing might be agnostic to the specific position at which it occurs.[46] By contrast, approximately equal numbers of genes (~2%–3%) have more or less cryptic splicing than expected in mature and CHX-treated RNA (Figure S6H). Genes with less cryptic splicing in mature RNA are enriched for genes involved in cytoplasmic translation, including those that encode for ribosomal protein genes.

Transcriptional variation influences cryptic splicing

The features most strongly associated with cryptic splicing in a gene increased opportunities for splicing to occur. Thus, we wondered if cryptic splicing was generally driven by mechanisms that increased transcriptional noise. To test this, we treated K562

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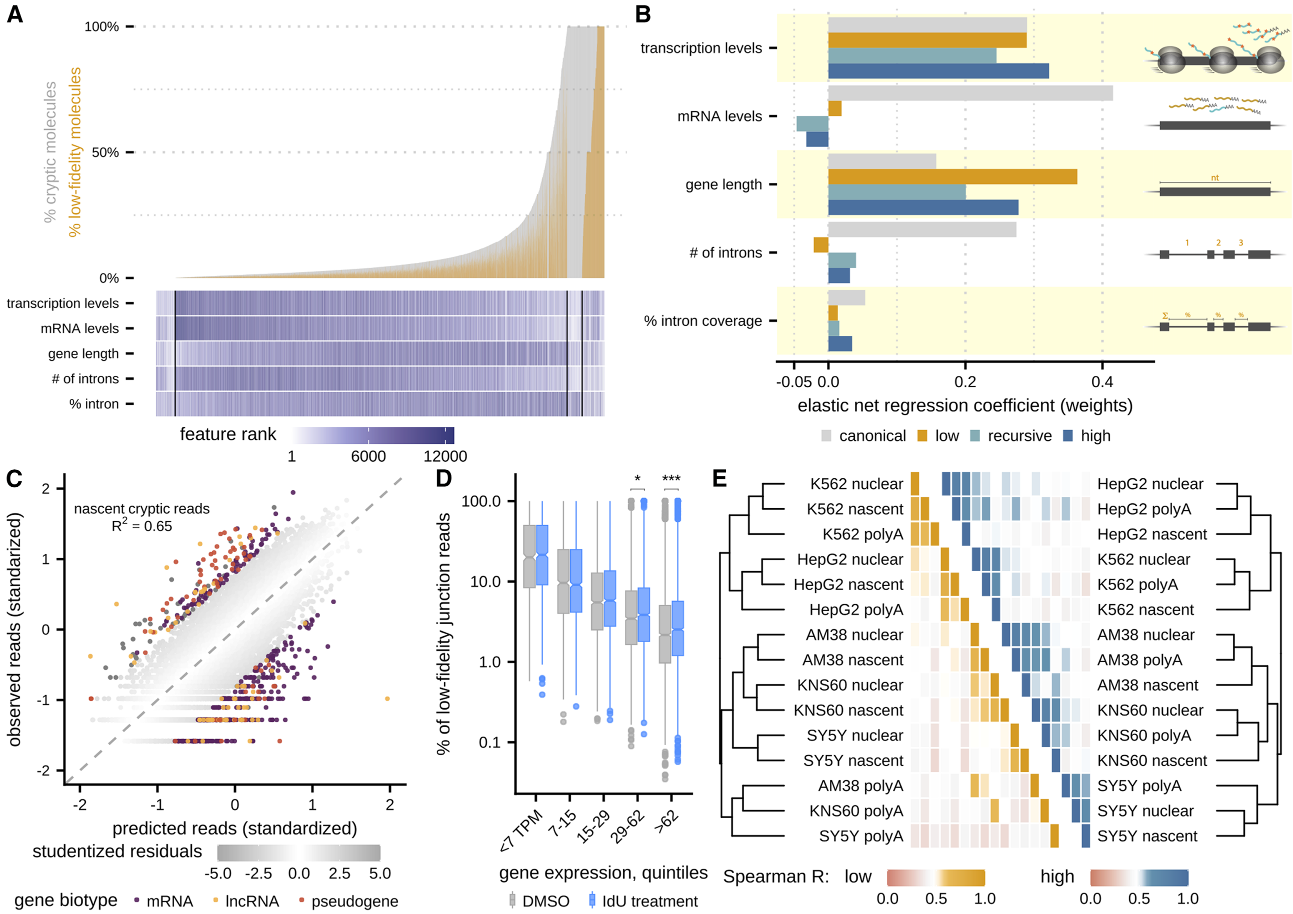

Figure 4. Cryptic splicing of a gene is associated with gene length and transcription levels

(A) Ranked percentage of transcripts with cryptic splicing (gray) and low-fidelity splicing (orange) per gene, using junction-specific TPMs. Below are associated transcription levels (nascent RNA TPMs), mRNA levels (mature RNA TPMs), gene length, number of total introns, and the percent of a gene annotated as intronic, where shading indicates rank of feature ordered from lowest to highest value.

(B) Feature weights from elastic net regression models trained on the number of cryptic reads per gene for canonical (gray), low-fidelity (orange), recursive (teal), and high-fidelity (dark blue) sites in nascent RNA.

(C) Predicted cryptic splicing per gene from elastic net regression (center standardized, *x* axis) versus observed cryptic splicing per gene in nascent RNA (*y* axis). Gray shading indicates studentized residuals, and genes with |studentized residual| $\geq$ 2 are colored by ENSEMBL biotype.

(D) Percentage of low-fidelity reads (relative to all junction reads) per gene in control (gray) and IdU-treated cells (blue) across bins of gene expression levels. Significance assessed with Mann-Whitney U test. * adjusted *p* value < 0.05. *** adjusted *p* value < 0.001.

(E) Clustering of samples across cell types and enrichment methods by extent of low-fidelity (left) or high-fidelity (right) splicing per gene. Dendrograms indicate hierarchical clustering distances, while heatmaps indicate pairwise Spearman correlations.

cells with 5′-iodo-2′-deoxyuridine (IdU)—a small molecule shown to increase transcriptional variance without affecting the mean of gene expression[48]—and collected nascent RNA. For highly expressed genes that allow for more cryptic site detection, we see significantly increased cryptic splicing, particularly among low-fidelity splice sites (Figure 4D). If transcriptional fluctuations were a primary driver of cryptic splicing across genes, we would also expect that genes should have similar extents of cryptic splicing across cell types after normalizing for RNA abundance. We generated nascent, nuclear, and mature RNA data from three additional cell lines—hepatocellular carcinoma cells (HepG2), neuroblastoma cells (SY5Y), and an additional glioblastoma line (AM38)—and evaluated cryptic splicing across cell types after normalizing for RNA abundance (STAR Methods). Samples tend to cluster first by cell type of origin and then RNA life stage (Figure 4E) for both low- and high-fidelity splicing. The two glioblastoma cell types (KNS60 and AM38) cluster together, following clustering of all neuronally derived cell types. The mature RNA samples from the neuronally derived cell types cluster apart from the nascent and nuclear samples, perhaps due to relatively low persistence of low-fidelity splicing in mature RNA. These results indicate that there are differences in either propensity for cryptic splicing or gene-specific tolerance for cryptic splicing across cell types.

Molecular determinants of cryptic splicing

To understand molecular influences on cryptic splicing levels, we first sought to confirm that cryptic sites are indeed due to

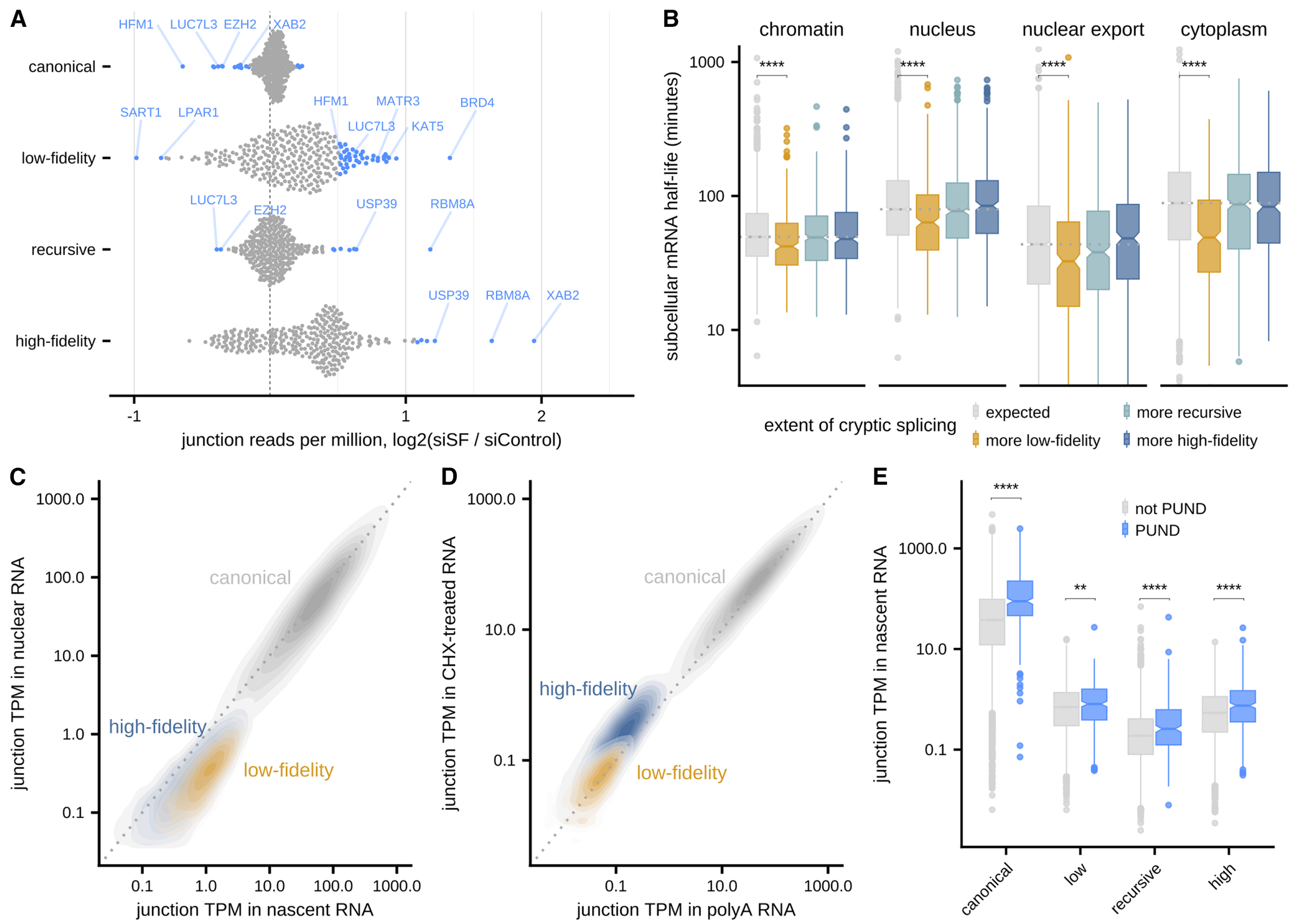

Figure 5. Cryptic transcripts are quickly cleared out of cellular compartments

(A) Fold change in junction reads between splicing-related protein (SF) KD and scrambled siRNA controls for canonical, low-fidelity, recursive, and high-fidelity junction reads across all 305 SFs from Rogalska et al.[52] Significance assessed with a permuted null distribution using control samples, and significantly changing factors are colored blue (adjusted p value < 0.05), with representative examples labeled.

(B) mRNA residence times or half-lives across cellular compartments estimated using subcellular TimeLapse-seq[57] for genes with expected (gray) or more low-fidelity (orange), recursive (teal), or high-fidelity (dark blue) transcripts than predicted by elastic net regression. Significance assessed with Mann-Whitney U test. **** adjusted p value < 0.0001.

(C) Junction-specific TPMs in nascent RNA (x axis) versus nuclear RNA (y axis).

(D) Junction-specific TPMs in mature RNA (x axis) versus CHX-treated mature RNA (y axis).

(E) Junction-specific TPMs across splice site categories for genes categorized as predicted to undergo nuclear degradation (PUNDs,[57] blue) or matched controls (gray). Significance assessed with Mann-Whitney U test. ** adjusted p value < 0.01. **** adjusted p value < 0.0001.

spliceosome activity. We assessed cryptic splicing following pharmacological depletion of spliceosome function by pladienolide B (Pla-B), which targets the core spliceosome component SF3B1 to inhibit U2 snRNP binding.[49,50] Using public nascent RNA data following Pla-B treatment in K562 cells,[51] we see fewer splice sites and junction reads across all splice site categories (Figure S7A). 84% and 79% of genes show reduced levels of low- and high-fidelity splicing, respectively, which is similar to the 88% of genes with reduced canonical splicing (Figure S7B). Among these genes, there is a 2.3-fold and 2.7-fold reduction, on average, in canonical and cryptic splicing, respectively. Notably, we do not see similar reductions in either canonical or cryptic splicing in total RNA, indicating that Pla-B treatment does not affect the clearance and degradation of mis-spliced transcripts. Together, these observations suggest cryptic sites are likely generated by spurious spliceosome activities.

We next sought to identify molecular factors that influence global cryptic splicing levels in cells. Using mature RNA-seq data after small interfering RNA (siRNA)-mediated knockdown (KD) of 305 splicing-related factors (SFs) from Rogalska et al.,[52] we evaluated significant changes in cryptic splicing observed in KD relative to scrambled siRNA controls (STAR Methods). We see that the majority of SF KDs result in global shifts toward more low-fidelity (72% of factors) and/or high-fidelity (76%) reads (Figure 5A). In addition to known splice site recognition factors (Table S6), SFs whose KD leads to significantly increased cryptic splicing include factors involved in chromatin modification and epigenetic regulation (i.e., BRD4 and KAT5), chromatin organization (i.e., MATR3), DNA damage and R-loop processing (i.e., KAT5 and XAB2), and mitotic checkpoints (i.e., USP39). SFs whose KD significantly change motif-consistent recursive site levels overlap both with factors that lead to more high-fidelity reads (i.e., USP39 and exon junction complex (EJC)-component RBM8A) and factors that decrease canonical reads (i.e., splice site recognition factor LUC7L3 and epigenetic regulator EZH2), again placing motif-consistent recursive sites at the intersection between regulated and cryptic sites. To cast a broader net, we also performed the same analysis using lower-depth RNA-seq data from siRNA-mediated KD of 204 RNA-binding proteins (RBPs) from the ENCODE consortium.[53] We see that 4 factors result in significant increases in motif-consistent recursive and high-fidelity cryptic reads and sites—SUPV3L1, EIF4A3, HNRNPC, and AQR (Figures S7D and S7E)—all of which are known to play a role in splicing regulation or quality control and some of which have been implicated in cryptic high-fidelity splice site usage.[54–56] These observations suggest that cryptic splicing might occur through a combination of favorable DNA (epigenetic marks, chromatin structure, or DNA damage repair) and RNA (canonical splicing regulators or post-transcriptional factors) environments. However, since both of these datasets are only derived from mature RNA, we are unable to distinguish between factors that may change the propensity for cryptic splice usage and factors involved in quality control or stability of cryptic transcripts.

The fate of cryptic transcripts may depend on splice site fidelity

Finally, we wanted to start understanding the ultimate cellular fate of transcripts with cryptic sites. Initial observations suggest that transcripts with cryptic sites—particularly those with low-fidelity sites—are likely pruned from nuclear and mature RNA pools. While high-fidelity cryptic splicing is consistent through RNA enrichments and more abundant after CHX treatment, low-fidelity sites substantially decrease in frequency through RNA life stages. To evaluate the lifetime of cryptic transcripts and cellular compartments in which they might be subject to regulation, we used subcellular residence times in K562 cells estimated using RNAFlow from Ietswaart et al.[57] We see that genes with more low-fidelity splicing than expected (from elastic net regression predictions) spend significantly less time on chromatin, have less time in the nucleus, have faster nuclear export rates, and are degraded faster in the cytoplasm (Figure 5B). One possibility is that these transcripts are generally unstable in every cellular compartment. However, it is also possible that transcripts using low-fidelity splice sites are recognized co-transcriptionally (leading to faster chromatin release; Figure S7F), quality control mechanisms lead to nuclear degradation, and thus there are fewer of these transcripts to export and degrade in the cytoplasm.

While challenging to distinguish between these two possibilities directly, we used the relative expression of cryptic transcripts across cellular compartments to indirectly ask about where degradation might be occurring. Transcripts with low-fidelity splice sites are expressed 3.3-fold more on average in nascent RNA than in total nuclear RNA, while high-fidelity cryptic transcripts are less shifted (2.4-fold) and canonical transcripts are highly correlated between the two factions (Pearson R = 0.91; 1.1-fold more in nascent RNA; Figure 5C). In mature RNA, high-fidelity cryptic transcripts are more affected by transcription inhibition with CHX (1.77-fold more expressed in CHX-treated RNA on average) than the expression of transcripts with canonical or low-fidelity sites (Pearson R = 0.97 and 0.96, respectively. Both 1.1-fold more in CHX-treated RNA; Figure 5D). These observations suggest a hypothesis in which many transcripts with cryptic sites, and especially those with low-fidelity splicing, may be recognized and degraded in the nucleus, while transcripts with high-fidelity splicing may be more likely to survive long enough to trigger cytoplasmic translation-dependent degradation mechanisms. Consistent with this, we see that the 340 genes predicted to undergo nuclear degradation (PUNDs) by RNAFlow analyses[57] have significantly more cryptic splicing in nascent RNA than non-PUND genes (Figure 5E). PUNDs also have more canonical transcripts as well, perhaps pointing to greater splicing activity in this class of genes.

DISCUSSION

Here, we systematically identify and quantify cryptic splicing across stages of the RNA lifecycle. Our results suggest widespread error by the mammalian splicing machinery, with unexplored roles for stochastic biological noise in shaping human transcriptome diversity. Previous measurements from steady-state RNA-seq data have estimated that ∼2% of transcription products might be erroneously spliced.[26] Our results indicate that this is likely an underestimate and that profiling splicing intermediates provides a more complete picture of cryptic splicing products.[4] By doing so, we are able to characterize the genomic features associated with 3 different categories of cryptic splicing: sites with high-fidelity, recursive, or low-fidelity spliceosome binding motifs. Since these categories were defined by dinucleotide sequences alone, future work using more nuanced classifications using full splice site scores or *in vivo* binding data could refine the characterization of cryptic splicing.

This work challenges the widespread perspective that all alternative RNA splicing events or isoforms are regulated and potentially have biological function. The advent of high-depth RNA-seq approaches has led to the identification of novel splicing events with high-fidelity spliceosome motifs but relatively low expression. Our results support the hypothesis put forward by previous modeling studies that the majority of these alternative isoforms arise from stochastic pairing between canonical splice sites.[15,58] Significant fluctuations in the levels of these products might be consequences of direct regulation of the major isoform, fluctuations in the factors regulating splice site usage, or degradation of unproductive isoforms. Consistent with a recent study that used nascent RNA-seq to study unproductive splicing in human cells,[33] we find that high-fidelity cryptic transcripts appear to be more stable through the RNA lifecycle and are likely targeted by translation-dependent degradation processes (Figure 5D). A subset of these sites, marked by recursive splice site motifs, might result in multi-step splicing of an intron. We identify 12%–13% of putative high-confidence human recursive sites previously identified in different cell lines and cellular contexts.[29,39] This is consistent with recent results that despite the

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presence of many sites with recursive motifs, bona fide recursive splicing happens rarely and stochastically in human cells[29] and suggests that usage of these sites more often results in unstable, unproductive transcripts.

The majority of cryptic sites we identify in newly created RNA use low-fidelity spliceosome motifs. These events must somehow bypass early kinetic proofreading steps, suggesting heterogeneity in initial spell-checking steps that evaluate surviving transcripts. This is consistent with the idea that kinetic proofreading cannot be perfect, at the risk of also reducing the production rate of productive transcripts if selection criteria are too stringent. Instead, there is likely a balance between productive and unproductive transcripts that survive proofreading steps. The existence of these cryptic sites points to an underappreciated role for stochasticity in the molecular interactions that underlie splicing mechanisms.[26] However, it remains unclear if there are genetic or biochemical factors that promote the usage of low-fidelity splice sites. These sites may be more representative of the sequence preferences of variant or modified snRNA.[59–61] Given that we observe specific dinucleotide pairs used more often as low-fidelity splice sites (Figures 3F and 3G), it is possible that local sequence composition promotes favorable spliceosomal structural conformations to allow intron excision. Similarly, local sequence composition may also influence RNA secondary or tertiary structures—such as G-quadruplexes[44] or base-pairing interactions between flanking regions—that create splicing-competent conformations.

Our results suggest that transcripts with low-fidelity splicing may be more likely to be turned over quickly in every cellular compartment, especially in the nucleus (Figures 5B and 5C). Given that the splicing motifs themselves are spliced out of final transcripts, the question remains: what leads to transcripts that use different cryptic sites to be flagged for degradation with variable spatio-temporal dynamics? We find that the KD of factors involved in chromatin regulation or DNA damage repair modulates global cryptic splicing (Figure 5A), allowing for the possibility that low-fidelity sites might be recognized (and potentially degraded) co-transcriptionally, perhaps prior to the resolution of R-loops by DNA repair machinery. Another possibility is that there are changes in or a lack of EJC deposition at exon-exon junctions mediated by low-fidelity splice sites. This could involve differential EJC composition at these sites, consistent with our observation that KD of known EJC components (i.e., EIF4A3) causes an increase in cryptic transcripts (Figures S7C and S7D). The sequence landscape around a junction might also influence EJC binding. A recent study found that m6A levels were depleted around canonical splice sites in pre-mRNA in a manner that enabled EJC deposition.[62] Consequently, m6A modification near cryptic sites may prevent EJC recruitment or deposition. More broadly, it is unclear precisely how transcripts with low-fidelity sites are degraded. Since the RNA exosome is known to be involved in nuclear degradation of erroneous RNA products,[22] this is the most obvious candidate for degrading cryptic transcripts in the nucleus. However, further experiments are needed to test and confirm this hypothesis. In particular, quantifying cryptic splicing following exosome component KDs or comparing to binding data from RNA decay factors would clarify whether rapid turnover reflects active surveillance or inherent instability of molecules with cryptic sites.

Finally, cryptic splicing is linked to increased genomic sequence space devoted to non-coding regions. This supports the multilevel selection theory that the prevalence of noisy splicing might be tolerated as a playground for evolutionary innovation even if only a small number of cryptic sites result in fitness advantages.[31,63] Low levels of cryptic events might benefit cells by allowing isoforms (and downstream products) to acquire new functions in stress, differentiation, or disease contexts. Thus, some waste of metabolic resources that creates mostly unproductive transcripts might not be completely onerous, depending on the function of the gene product or cellular context. This is consistent with our observation that genes involved in core cellular functions (i.e., cytoplasmic translation) tolerate less cryptic splicing, while lncRNA and expressed pseudogenes have less precise splicing regulation (Figure 4C; Figure S6H). It is unclear whether these constraints occur at the level of splice site selection, quality control, or degradation and how specific genes are differentially regulated. We see cell-type-specific variability in the abundance of cryptic site usage across genes—even for low-fidelity sites—suggesting that certain cellular environments might be more or less permissive for splicing noise. Cryptic splicing might be even more prevalent in cellular contexts or diseases such as cancer, where transcriptional noise could tax the splicing machinery and result in more chances for splicing error.[64,65] Our results have established substantial variability in cryptic site usage across genes and cells, but more work is necessary to understand the causes, consequences, and landscape of cryptic splicing in mammalian cells.

Limitations of the study

To identify cryptic splice sites, we developed a computational framework called *CRYPTID-SS*. Within this workflow, we tried to account for (and remove) many artifactual sources of cryptic junction reads—including transposon or sequence variability, biases created during RNA handling or library preparation, and sequencing or mapping errors—using control datasets to benchmark the probability that these issues may arise. It is possible that other unknown biological or technical causes of spurious junction reads may exist, which may affect the confidence in the positioning or quantification of any given lowly expressed cryptic site. However, any such biases should be consistent across genes and/or RNA enrichment types and have minimal effects on our results from analyses of aggregate genomic patterns.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Athma A. Pai (athma.pai@umassmed.edu).

Materials availability

All unique/stable reagents generated in this study are available from the lead contact with a completed materials transfer agreement.

Data and code availability

- All sequencing data generated for this study are available at the Gene Expression Omnibus (GSE302375).
- The *CRYPTID-SS* pipeline and code for downstream analyses included in this manuscript are available at https://github.com/thepailab/CRYPTID-SS, deposited at Zenodo, and are publicly available at https://doi.org/10.5281/zenodo.21197326 as of the date of publication.

- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

E.S.K. and K.B. contributed equally. J.K.W. and A.A.P. conceived and designed experiments. K.B., Z.J.K., Y.L., V.S., and N.J. performed the experiments. E.S.K., Y.L., A.K., and A.A.P. performed statistical analyses. E.S.K., Y.L., A.K., and A.A.P. analyzed the data. J.K.W., E.C.R., and A.A.P. contributed reagents/materials/analysis tools. E.S.K., K.B., and A.A.P. wrote the paper. J.K.W. and A.A.P. jointly supervised research.

DECLARATION OF INTERESTS

A.A.P., J.K.W., K.B., E.S.K., and Z.J.K. have a pending patent application (provisional application number: US20230022489A1).

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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REFERENCES

1. Hollander, D., Naftelberg, S., Lev-Maor, G., Kornblihtt, A.R., and Ast, G. (2016). How Are Short Exons Flanked by Long Introns Defined and Committed to Splicing? Trends Genet. *32*, 596–606. https://doi.org/10.1016/j.tig.2016.07.003.
2. Saudemont, B., Popa, A., Parmley, J.L., Rocher, V., Blugeon, C., Necsulea, A., Meyer, E., and Duret, L. (2017). The fitness cost of mis-splicing is the main determinant of alternative splicing patterns. Genome Biol. *18*, 208. https://doi.org/10.1186/s13059-017-1344-6.
3. Minkin, I., and Salzberg, S.L. (2025). Conservation assessment of human splice site annotation based on a 470-genome alignment. Nucleic Acids Res. *53*, gkaf184. https://doi.org/10.1093/nar/gkaf184.
4. Wan, Y., and Larson, D.R. (2018). Splicing heterogeneity: separating signal from noise. Genome Biol. *19*, 86. https://doi.org/10.1186/s13059-018-1467-4.
5. Rogalska, M.E., Vivori, C., and Valcárcel, J. (2023). Regulation of pre-mRNA splicing: roles in physiology and disease, and therapeutic prospects. Nat. Rev. Genet. *24*, 251–269. https://doi.org/10.1038/s41576-022-00556-8.
6. Wilkinson, M.E., Charenton, C., and Nagai, K. (2020). RNA splicing by the spliceosome. Annu. Rev. Biochem. *89*, 359–388. https://doi.org/10.1146/annurev-biochem-091719-064225.
7. Dvinge, H. (2018). Regulation of alternative mRNA splicing: old players and new perspectives. FEBS Lett. *592*, 2987–3006. https://doi.org/10.1002/1873-3468.13119.
8. Marasco, L.E., and Kornblihtt, A.R. (2023). The physiology of alternative splicing. Nat. Rev. Mol. Cell Biol. *24*, 242–254. https://doi.org/10.1038/s41580-022-00545-z.
9. Shenasa, H., and Bentley, D.L. (2023). Pre-mRNA splicing and its cotranscriptional connections. Trends Genet. *39*, 672–685. https://doi.org/10.1016/j.tig.2023.04.008.
10. Carrocci, T.J., and Neugebauer, K.M. (2024). Emerging and re-emerging themes in co-transcriptional pre-mRNA splicing. Mol. Cell *84*, 3656–3666. https://doi.org/10.1016/j.molcel.2024.08.036.
11. Carey, L.B. (2015). RNA polymerase errors cause splicing defects and can be regulated by differential expression of RNA polymerase subunits. Elife *4*, e09945. https://doi.org/10.7554/eLife.09945.
12. Gout, J.-F., Li, W., Fritsch, C., Li, A., Haroon, S., Singh, L., Hua, D., Fazelinia, H., Smith, Z., Seeholzer, S., et al. (2017). The landscape of transcription errors in eukaryotic cells. Sci. Adv. *3*, e1701484. https://doi.org/10.1126/sciadv.1701484.
13. Mohler, K., and Ibba, M. (2017). Translational fidelity and mistranslation in the cellular response to stress. Nat. Microbiol. *2*, 17117. https://doi.org/10.1038/nmicrobiol.2017.117.
14. Fox-Walsh, K.L., and Hertel, K.J. (2009). Splice-site pairing is an intrinsically high fidelity process. Proc. Natl. Acad. Sci. USA *106*, 1766–1771. https://doi.org/10.1073/pnas.0813128106.
15. Melamud, E., and Moult, J. (2009). Stochastic noise in splicing machinery. Nucleic Acids Res. *37*, 4873–4886. https://doi.org/10.1093/nar/gkp471.
16. Semlow, D.R., and Staley, J.P. (2012). Staying on message: ensuring fidelity in pre-mRNA splicing. Trends Biochem. Sci. *37*, 263–273. https://doi.org/10.1016/j.tibs.2012.04.001.
17. Skandalis, A. (2016). Estimation of the minimum mRNA splicing error rate in vertebrates. Mutat. Res. *784–785*, 34–38. https://doi.org/10.1016/j.mrfmmm.2016.01.002.
18. Davidson, L., Kerr, A., and West, S. (2012). Co-transcriptional degradation of aberrant pre-mRNA by Xrn2. EMBO J. *31*, 2566–2578. https://doi.org/10.1038/emboj.2012.101.
19. Egecioglu, D.E., and Chanfreau, G. (2011). Proofreading and spellchecking: a two-tier strategy for pre-mRNA splicing quality control. RNA *17*, 383–389. https://doi.org/10.1261/rna.2454711.

20. Thomas, M.J., Platas, A.A., and Hawley, D.K. (1998). Transcriptional fidelity and proofreading by RNA polymerase II. Cell *93*, 627–637. https://doi.org/10.1016/S0092-8674(00)81191-5.

21. Sydow, J.F., and Cramer, P. (2009). RNA polymerase fidelity and transcriptional proofreading. Curr. Opin. Struct. Biol. *19*, 732–739. https://doi.org/10.1016/j.sbi.2009.10.009.

22. Schmid, M., and Jensen, T.H. (2018). Controlling nuclear RNA levels. Nat. Rev. Genet. *19*, 518–529. https://doi.org/10.1038/s41576-018-0013-2.

23. Morris, C., Cluet, D., and Ricci, E.P. (2021). Ribosome dynamics and mRNA turnover, a complex relationship under constant cellular scrutiny. Wiley Interdiscip. Rev., RNA *12*, e1658. https://doi.org/10.1002/wrna.1658.

24. Monaghan, L., Longman, D., and Cáceres, J.F. (2023). Translation-coupled mRNA quality control mechanisms. EMBO J. *42*, e114378. https://doi.org/10.15252/embj.2023114378.

25. Dou, Y., Fox-Walsh, K.L., Baldi, P.F., and Hertel, K.J. (2006). Genomic splice-site analysis reveals frequent alternative splicing close to the dominant splice site. RNA *12*, 2047–2056. https://doi.org/10.1261/rna.151106.

26. Pickrell, J.K., Pai, A.A., Gilad, Y., and Pritchard, J.K. (2010). Noisy Splicing Drives mRNA Isoform Diversity in Human Cells. PLOS Genet. *6*, e1001236. https://doi.org/10.1371/journal.pgen.1001236.

27. Hatton, A.R., Subramaniam, V., and Lopez, A.J. (1998). Generation of alternative Ultrabithorax isoforms and stepwise removal of a large intron by resplicing at exon-exon junctions. Mol. Cell *2*, 787–796. https://doi.org/10.1016/S1097-2765(00)80293-2.

28. Burnette, J.M., Miyamoto-Sato, E., Schaub, M.A., Conklin, J., and Lopez, A.J. (2005). Subdivision of large introns in Drosophila by recursive splicing at nonexonic elements. Genetics *170*, 661–674. https://doi.org/10.1534/genetics.104.039701.

29. Wan, Y., Anastasakis, D.G., Rodriguez, J., Palangat, M., Gudla, P., Zaki, G., Tandon, M., Pegoraro, G., Chow, C.C., Hafner, M., et al. (2021). Dynamic imaging of nascent RNA reveals general principles of transcription dynamics and stochastic splice site selection. Cell *184*, 2878–2895.e20. https://doi.org/10.1016/j.cell.2021.04.012.

30. Pai, A.A., Paggi, J.M., Yan, P., Adelman, K., and Burge, C.B. (2018). Numerous recursive sites contribute to accuracy of splicing in long introns in flies. PLOS Genet. *14*, e1007588. https://doi.org/10.1371/journal.pgen.1007588.

31. Sibley, C.R., Blazquez, L., and Ule, J. (2016). Lessons from non-canonical splicing. Nat. Rev. Genet. *17*, 407–421. https://doi.org/10.1038/nrg.2016.46.

32. Zhang, Z., Xin, D., Wang, P., Zhou, L., Hu, L., Kong, X., and Hurst, L.D. (2009). Noisy splicing, more than expression regulation, explains why some exons are subject to nonsense-mediated mRNA decay. BMC Biol. *7*, 23. https://doi.org/10.1186/1741-7007-7-23.

33. Fair, B., Buen Abad Najar, C.F., Zhao, J., Lozano, S., Reilly, A., Mossian, G., Staley, J.P., Wang, J., and Li, Y.I. (2024). Global impact of unproductive splicing on human gene expression. Nat. Genet. *56*, 1851–1861. https://doi.org/10.1038/s41588-024-01872-x.

34. McManus, C.J., Duff, M.O., Eipper-Mains, J., and Graveley, B.R. (2010). Global analysis of trans-splicing in Drosophila. Proc. Natl. Acad. Sci. USA *107*, 12975–12979. https://doi.org/10.1073/pnas.1007586107.

35. Lim, K.H., Han, Z., Jeon, H.Y., Kach, J., Jing, E., Weyn-Vanhentenryck, S., Downs, M., Corrionero, A., Oh, R., Scharner, J., et al. (2020). Antisense oligonucleotide modulation of non-productive alternative splicing upregulates gene expression. Nat. Commun. *11*, 3501. https://doi.org/10.1038/s41467-020-17093-9.

36. Bradnam, K.R., and Korf, I. (2008). Longer first introns are a general property of eukaryotic gene structure. PLOS One *3*, e3093. https://doi.org/10.1371/journal.pone.0003093.

37. Duff, M.O., Olson, S., Wei, X., Garrett, S.C., Osman, A., Bolisetty, M., Plocik, A., Celniker, S.E., and Graveley, B.R. (2015). Genome-wide identification of zero nucleotide recursive splicing in Drosophila. Nature *521*, 376–379. https://doi.org/10.1038/nature14475.

38. Sibley, C.R., Emmett, W., Blazquez, L., Faro, A., Haberman, N., Briese, M., Trabzuni, D., Ryten, M., Weale, M.E., Hardy, J., et al. (2015). Recursive splicing in long vertebrate genes. Nature *521*, 371–375. https://doi.org/10.1038/nature14466.

39. Zhang, X.-O., Fu, Y., Mou, H., Xue, W., and Weng, Z. (2018). The temporal landscape of recursive splicing during Pol II transcription elongation in human cells. PLOS Genet. *14*, e1007579. https://doi.org/10.1371/journal.pgen.1007579.

40. Hoppe, E.R., Udy, D.B., and Bradley, R.K. (2023). Recursive splicing discovery using lariats in total RNA sequencing. Life Sci. Alliance *6*, e202201889. https://doi.org/10.26508/lsa.202201889.

41. Bradley, R.K., Merkin, J., Lambert, N.J., and Burge, C.B. (2012). Alternative splicing of RNA triplets is often regulated and accelerates proteome evolution. PLOS Biol. *10*, e1001229. https://doi.org/10.1371/journal.pbio.1001229.

42. Xia, X. (2012). Position weight matrix, gibbs sampler, and the associated significance tests in motif characterization and prediction. Scientifica (Cairo) *2012*, 917540. https://doi.org/10.6064/2012/917540.

43. Capitanchik, C., Wilkins, O.G., Wagner, N., Gagneur, J., and Ule, J. (2025). From computational models of the splicing code to regulatory mechanisms and therapeutic implications. Nat. Rev. Genet. *26*, 171–190. https://doi.org/10.1038/s41576-024-00774-2.

44. Georgakopoulos-Soares, I., Parada, G.E., Wong, H.Y., Medhi, R., Furlan, G., Munita, R., Miska, E.A., Kwok, C.K., and Hemberg, M. (2022). Alternative splicing modulation by G-quadruplexes. Nat. Commun. *13*, 2404. https://doi.org/10.1038/s41467-022-30071-7.

45. Sheth, N., Roca, X., Hastings, M.L., Roeder, T., Krainer, A.R., and Sachidanandam, R. (2006). Comprehensive splice-site analysis using comparative genomics. Nucleic Acids Res. *34*, 3955–3967. https://doi.org/10.1093/nar/gkl556.

46. Staněk, D. (2021). Long non-coding RNAs and splicing. Essays Biochem. *65*, 723–729. https://doi.org/10.1042/EBC20200087.

47. Basu, K., Dey, A., and Kiran, M. (2023). Inefficient splicing of long non-coding RNAs is associated with higher transcript complexity in human and mouse. RNA Biol. *20*, 563–572. https://doi.org/10.1080/15476286.2023.2242649.

48. Desai, R.V., Chen, X., Martin, B., Chaturvedi, S., Hwang, D.W., Li, W., Yu, C., Ding, S., Thomson, M., Singer, R.H., et al. (2021). A DNA repair pathway can regulate transcriptional noise to promote cell fate transitions. Science *373*, eabc6506. https://doi.org/10.1126/science.abc6506.

49. Kotake, Y., Sagane, K., Owa, T., Mimori-Kiyosue, Y., Shimizu, H., Uesugi, M., Ishihama, Y., Iwata, M., and Mizui, Y. (2007). Splicing factor SF3b as a target of the antitumor natural product pladienolide. Nat. Chem. Biol. *3*, 570–575. https://doi.org/10.1038/nchembio.2007.16.

50. Effenberger, K.A., Urabe, V.K., Prichard, B.E., Ghosh, A.K., and Jurica, M.S. (2016). Interchangeable SF3B1 inhibitors interfere with pre-mRNA splicing at multiple stages. RNA *22*, 350–359. https://doi.org/10.1261/rna.053108.115.

51. Caizzi, L., Monteiro-Martins, S., Schwalb, B., Lysakovskaia, K., Schmitzova, J., Sawicka, A., Chen, Y., Lidschreiber, M., and Cramer, P. (2021). Efficient RNA polymerase II pause release requires U2 snRNP function. Mol. Cell *81*, 1920–1934.e9. https://doi.org/10.1016/j.molcel.2021.02.016.

52. Rogalska, M.E., Mancini, E., Bonnal, S., Gohr, A., Dunyak, B.M., Arecco, N., Smith, P.G., Vaillancourt, F.H., and Valcárcel, J. (2024). Transcriptome-wide splicing network reveals specialized regulatory functions of the core spliceosome. Science *386*, 551–560. https://doi.org/10.1126/science.adn8105.

53. ENCODE; Project Consortium (2012). An integrated encyclopedia of DNA elements in the human genome. Nature *489*, 57–74. https://doi.org/10.1038/nature11247.

54. Feng, Q., Krick, K., Chu, J., and Burge, C.B. (2023). Splicing quality control mediated by DHX15 and its G-patch activator SUGP1. Cell Rep. *42*, 113223. https://doi.org/10.1016/j.celrep.2023.113223.

55. García-Ruiz, S., Zhang, D., Gustavsson, E.K., Rocamora-Perez, G., Grant-Peters, M., Fairbrother-Browne, A., Reynolds, R.H., Brenton, J.W., Gil-Martínez, A.L., Chen, Z., et al. (2025). Splicing accuracy varies across human introns, tissues, age and disease. Nat. Comm. *16*, 1068. https://doi.org/10.1038/s41467-024-55607.

56. Van Nostrand, E.L., Pratt, G.A., Yee, B.A., Wheeler, E.C., Blue, S.M., Mueller, J., Park, S.S., Garcia, K.E., Gelboin-Burkhart, C., Nguyen, T.B., et al. (2020). Principles of RNA processing from analysis of enhanced CLIP maps for 150 RNA binding proteins. Genome Biol. *21*, 90. https://doi.org/10.1186/s13059-020-01982-9.

57. Ietswaart, R., Smalec, B.M., Xu, A., Choquet, K., McShane, E., Jowhar, Z.M., Guegler, C.K., Baxter-Koenigs, A.R., West, E.R., Fu, B.X.H., et al. (2024). Genome-wide quantification of RNA flow across subcellular compartments reveals determinants of the mammalian transcript life cycle. Mol. Cell *84*, 2765–2784.e16. https://doi.org/10.1016/j.molcel.2024.06.008.

58. Hu, J., Boritz, E., Wylie, W., and Douek, D.C. (2017). Stochastic principles governing alternative splicing of RNA. PLOS Comput. Biol. *13*, e1005761. https://doi.org/10.1371/journal.pcbi.1005761.

59. Mabin, J.W., Lewis, P.W., Brow, D.A., and Dvinge, H. (2021). Human spliceosomal snRNA sequence variants generate variant spliceosomes. RNA *27*, 1186–1203. https://doi.org/10.1261/rna.078768.121.

60. Parker, M.T., Fica, S.M., Barton, G.J., and Simpson, G.G. (2023). Inter-species association mapping links splice site evolution to METTL16 and SNRNP27K. Elife *12*, e91997. https://doi.org/10.7554/eLife.91997.

61. Parker, M.T., Soanes, B.K., Kusakina, J., Larrieu, A., Knop, K., Joy, N., Breidenbach, F., Sherwood, A.V., Barton, G.J., Fica, S.M., et al. (2022). m6A modification of U6 snRNA modulates usage of two major classes of pre-mRNA 5′ splice site. Elife *11*, e78808. https://doi.org/10.7554/eLife.78808.

62. Sethi, A.J., Guarnacci, M., Bilal, M., Krishnan, K.S., Hayashi, A., Kanchi, M., Nojima, T., Preiss, T., Eyras, E., and Hayashi, R. (2025). Single-molecule multimodal timing of*in vivo*mRNA synthesis. Preprint at bioRxiv. https://doi.org/10.1101/2025.04.27.650906.

63. Brunet, T.D.P., and Doolittle, W.F. (2015). Multilevel Selection Theory and the Evolutionary Functions of Transposable Elements. Genome Biol. Evol. *7*, 2445–2457. https://doi.org/10.1093/gbe/evv152.

64. Lin, C.Y., Lovén, J., Rahl, P.B., Paranal, R.M., Burge, C.B., Bradner, J.E., Lee, T.I., and Young, R.A. (2012). Transcriptional amplification in tumor cells with elevated c-Myc. Cell *151*, 56–67. https://doi.org/10.1016/j.cell.2012.08.026.

65. Chen, L., Tovar-Corona, J.M., and Urrutia, A.O. (2011). Increased levels of noisy splicing in cancers, but not for oncogene-derived transcripts. Hum. Mol. Genet. *20*, 4422–4429. https://doi.org/10.1093/hmg/ddr370.

66. Quentmeier, H., Pommerenke, C., Dirks, W.G., Eberth, S., Koeppel, M., MacLeod, R.A.F., Nagel, S., Steube, K., Uphoff, C.C., and Drexler, H.G. (2019). The LL-100 panel: 100 cell lines for blood cancer studies. Sci. Rep. *9*, 8218. https://doi.org/10.1038/s41598-019-44491-x.

67. Bolger, A.M., Lohse, M., and Usadel, B. (2014). Trimmomatic: a flexible trimmer for Illumina sequence data. Bioinformatics *30*, 2114–2120. https://doi.org/10.1093/bioinformatics/btu170.

68. Zhang, J., Kobert, K., Flouri, T., and Stamatakis, A. (2014). PEAR: a fast and accurate Illumina Paired-End reAd mergeR. Bioinformatics *30*, 614–620. https://doi.org/10.1093/bioinformatics/btt593.

69. Xue, Z. gtf2csv: Convert Genome Annotation GTF File into Plain CSV Format (GitHub). https://github.com/zyxue/gtf2csv.

70. Bray, N.L., Pimentel, H., Melsted, P., and Pachter, L. (2016). Near-optimal probabilistic RNA-seq quantification. Nat. Biotechnol. *34*, 525–527. https://doi.org/10.1038/nbt.3519.

71. Quinlan, A.R., and Hall, I.M. (2010). BEDTools: a flexible suite of utilities for comparing genomic features. Bioinformatics *26*, 841–842. https://doi.org/10.1093/bioinformatics/btq033.

72. Dobin, A., Davis, C.A., Schlesinger, F., Drenkow, J., Zaleski, C., Jha, S., Batut, P., Chaisson, M., and Gingeras, T.R. (2013). STAR: ultrafast universal RNA-seq aligner. Bioinformatics *29*, 15–21. https://doi.org/10.1093/bioinformatics/bts635.

73. Van der Auwera, G.A., Carneiro, M.O., Hartl, C., Poplin, R., Del Angel, G., Levy-Moonshine, A., Jordan, T., Shakir, K., Roazen, D., Thibault, J., et al. (2013). From FastQ data to high confidence variant calls: the Genome Analysis Toolkit best practices pipeline. Curr. Protoc. Bioinform. *43*, 11.10.1–11.10.33. https://doi.org/10.1002/0471250953.bi1110s43.

74. Nabbi, A., and Riabowol, K. (2015). Rapid isolation of nuclei from cells in vitro. Cold Spring Harb. Protoc. *2015*, 769–772. https://doi.org/10.1101/pdb.prot083733.

75. Garibaldi, A., Carranza, F., and Hertel, K.J. (2017). Isolation of newly transcribed RNA using the metabolic label 4-thiouridine. Methods Mol. Biol. *1648*, 169–176. https://doi.org/10.1007/978-1-4939-7204-3_13.

76. Duffy, E.E., and Simon, M.D. (2016). Enriching s4 U-RNA using methane thiosulfonate (MTS) chemistry. Curr. Protoc. Chem. Biol. *8*, 234–250. https://doi.org/10.1002/cpch.12.

77. Kõressaar, T., Lepamets, M., Kaplinski, L., Raime, K., Andreson, R., and Remm, M. (2018). Primer3_masker: integrating masking of template sequence with primer design software. Bioinformatics *34*, 1937–1938. https://doi.org/10.1093/bioinformatics/bty036.

78. Untergasser, A., Cutcutache, I., Koressaar, T., Ye, J., Faircloth, B.C., Remm, M., and Rozen, S.G. (2012). Primer3–new capabilities and interfaces. Nucleic Acids Res. *40*, e115. https://doi.org/10.1093/nar/gks596.

79. Andrews, S. (2016). FastQC: A quality control tool for high throughput sequence data. https://www.bioinformatics.babraham.ac.uk/projects/fastqc/.

80. Li, H. (2018). seqtk. seqtk: Toolkit for processing sequences in FASTA/Q formats. https://github.com/lh3/seqtk.

81. Yeo, G., and Burge, C.B. (2004). Maximum entropy modeling of short sequence motifs with applications to RNA splicing signals. J. Comput. Biol. *11*, 377–394. https://doi.org/10.1089/1066527041410418.

82. Siepel, A., Bejerano, G., Pedersen, J.S., Hinrichs, A.S., Hou, M., Rosenbloom, K., Clawson, H., Spieth, J., Hillier, L.W., Richards, S., et al. (2005). Evolutionarily conserved elements in vertebrate, insect, worm, and yeast genomes. Genome Res. *15*, 1034–1050. https://doi.org/10.1101/gr.3715005.

83. Raney, B.J., Barber, G.P., Benet-Pagès, A., Casper, J., Clawson, H., Cline, M.S., Diekhans, M., Fischer, C., Navarro Gonzalez, J., Hickey, G., et al. (2024). The UCSC Genome Browser database: 2024 update. Nucleic Acids Res. *52*, D1082–D1088. https://doi.org/10.1093/nar/gkad987.

84. Picard, Toolkit. Broad Institute, GitHub Repository. https://broadinstitute.github.io/picard.

85. Poplin, R., Ruano-Rubio, V., DePristo, M.A., Fennell, T.J., Carneiro, M.O., Van der Auwera, G.A., Kling, D.E., Gauthier, L.D., Levy-Moonshine, A., Roazen, D., et al. (2017). Scaling accurate genetic variant discovery to tens of thousands of samples. Preprint at bioRxiv. https://doi.org/10.1101/201178.

86. Calvo-Roitberg, E., Lehman, J.W., Tam, E., Elhajjajy, S., Engelhardt, B.E., and Pai, A.A. (2025). SPARK: in silico simulations for benchmarking nascent RNA sequencing experiments. Preprint at bioRxiv. https://doi.org/10.1101/2025.11.06.687082.

87. Li, H., Handsaker, B., Wysoker, A., Fennell, T., Ruan, J., Homer, N., Marth, G., Abecasis, G., and Durbin, R.; 1000 Genome Project Data Processing Subgroup (2009). The Sequence Alignment/Map format and SAMtools. Bioinformatics *25*, 2078–2079. https://doi.org/10.1093/bioinformatics/btp352.

88. Kent, W.J. (2002). BLAT—the BLAST-like alignment tool. Genome Res. *12*, 656–664. https://doi.org/10.1101/gr.229202.

89. Groemping, U. (2006). Relative importance for linear regression in R: The package relaimpo. J. Stat. Softw. *17*, 1–27.

90. Tilgner, H., Knowles, D.G., Johnson, R., Davis, C.A., Chakrabortty, S., Djebali, S., Curado, J., Snyder, M., Gingeras, T.R., and Guigó, R. (2012). Deep sequencing of subcellular RNA fractions shows splicing to be predominantly co-transcriptional in the human genome but inefficient for lncRNAs. Genome Res. *22*, 1616–1625. https://doi.org/10.1101/gr.134445.111.

91. Herzel, L., and Neugebauer, K.M. (2015). Quantification of co-transcriptional splicing from RNA-Seq data. Methods *85*, 36–43. https://doi.org/10.1016/j.ymeth.2015.04.024.

92. Bailey, T.L., Johnson, J., Grant, C.E., and Noble, W.S. (2015). The MEME suite. Nucleic Acids Res. *43*, W39–W49. https://doi.org/10.1093/nar/gkv416.

93. Lim, L.P., and Burge, C.B. (2001). A computational analysis of sequence features involved in recognition of short introns. Proc. Natl. Acad. Sci. USA *98*, 11193–11198. https://doi.org/10.1073/pnas.201407298.

94. Wang, Z., Rolish, M.E., Yeo, G., Tung, V., Mawson, M., and Burge, C.B. (2004). Systematic identification and analysis of exonic splicing silencers. Cell *119*, 831–845. https://doi.org/10.1016/j.cell.2004.11.010.

95. Wang, Y., Ma, M., Xiao, X., and Wang, Z. (2012). Intronic splicing enhancers, cognate splicing factors and context-dependent regulation rules. Nat. Struct. Mol. Biol. *19*, 1044–1052. https://doi.org/10.1038/nsmb.2377.

96. Wang, Y., and Wang, Z. (2014). Systematical identification of splicing regulatory cis-elements and cognate trans-factors. Methods *65*, 350–358. https://doi.org/10.1016/j.ymeth.2013.08.019.

97. Fairbrother, W.G., Yeh, R.-F., Sharp, P.A., and Burge, C.B. (2002). Predictive identification of exonic splicing enhancers in human genes. Science *297*, 1007–1013. https://doi.org/10.1126/science.1073774.

98. Kuhn, M. (2008). Building Predictive Models inRUsing thecaretPackage. J. Stat. Softw. *28*, 1–26.

99. Elizarraras, J.M., Liao, Y., Shi, Z., Zhu, Q., Pico, A.R., and Zhang, B. (2024). WebGestalt 2024: faster gene set analysis and new support for metabolomics and multi-omics. Nucleic Acids Res. *52*, W415–W421. https://doi.org/10.1093/nar/gkae456.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
anti-vinculin (polyclonal rabbit antibody)	Cell Signaling Technologies	Cat. # 4650; RRID: AB_10559207
anti-β-actin (polyclonal rabbit antibody)	Cell Signaling Technologies	Cat. # 4967; RRID: AB_330288
anti-rabbit secondary (HRP-linked goat antibody)	Cell Signaling Technologies	Cat. # 7074; RRID: AB_2099233
Chemicals, peptides, and recombinant proteins		
Cycloheximide	Sigma	Cat#C7698
NMDi14	Sigma	Cat#SML1538
5-Iodo-2′deoxyuridine	Sigma	Cat#17125-5G
4sU	Sigma	Cat#T4509
Roche EDTA-free protease inhibitor	Millipora Sigma	Cat#11836153001
Sigma PhosSTOP phosphatase inhibitor	Millipore Sigma	Cat#4906845001
MTSEA biotin-XX	VWR	Cat#89139-636
Uridine	ChemGenes	Cat#RP-1186
Critical commercial assays		
KAPA Stranded mRNA-Seq Kit for Illumina Platforms	KAPA	Cat#KK8421
KAPA RNA HyperPrep Kit with RiboErase	KAPA	Cat#KK8561
Luna Universal qPCR Master Mix	NEB	Cat#M3003
Deposited data		
Raw and analyzed RNA-seq data	This paper	GSE302375
K562 whole genome sequencing data	ENCODE Consortium[53]	ENCSR045NDZ
K562 whole exome sequencing data	Quentmeier et al.[66]	ERR2990087
RNA-seq and TT-seq in K562 cells following PlaB treatment	Caizzi et al.[51]	GSE148433
RNA-seq following shRNA knockdown of splicing associated genes in HeLA cells	Rogalska et al.[52]	E-MTAB-11202
RNA-seq following shRNA knockdown of RNA binding protein genes in K562 cells	ENCODE Consortium[53]	www.encodeproject.org/
Experimental models: Cell lines		
AM-38 cells	JCRB Cell Bank (NIBN Japan)	Cat.# IFO50492
HepG2 cells	ATCC	Cat.# HB-8065
K562 cells	ATCC	Cat.# CCL243
KNS-60 cells	JCRB Cell Bank (NIBN Japan)	Cat.# IFO50357
SH-SY5Y cells	ATCC	Cat.# CRL2266
Oligonucleotides		
SIRV-Set1	Lexogen	Cat#0.25.03
SIRV-Set4	Lexogen	Cat#141.01
Software and algorithms		
Trimmomatic v0.32	Bolger et al.[67]	http://www.usadellab.org/cms/?page=trimmomatic
PEAR v0.9.11	Zhang et al.[68]	https://cme.h-its.org/exelixis/web/software/pear/doc.html
gtf2csv c0.2	Xue[69]	https://github.com/zyxue/gtf2csv
Kallisto v0.45.0	Bray et al.[70]	https://pachterlab.github.io/kallisto/download

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Bedtools v2.28.0	Quinlan and Hall[71]	https://bedtools.readthedocs.io/en/latest/content/installation.html
STAR v2.7.0e	Dobin et al.[72]	https://github.com/alexdobin/STAR
GATK v4.3.0.0	Van der Auwera et al.[73]	https://github.com/broadinstitute/gatk
CRYPTID-SS	This paper	https://github.com/thepailab/CRYPTID-SS https://doi.org/10.5281/zenodo.21197326

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Human cell culture

K562 cells were grown in IMDM supplemented with 10% FBS. KNS60 cells were grown in DMEM supplemented with 5% FBS. AM38 cells were grown in EMEM supplemented with 20% FBS. HepG2 cells were grown in DMEM supplemented with 10% FBS. SH-SY5Y cells were grown in DMEM:F12 supplemented with 10% FBS. All cell lines were maintained at 37°C and 5% CO_2 until 70% confluent. For cycloheximide experiments, K562 or KNS60 cells were treated 50ug/mL cycloheximide (CHX) (Sigma, C7698) resuspended in DMSO for 3 hours (or only DMSO as a control). For NMDi14 experiments, K562 and KNS60 cells were treated with 50uM NMDi14 (Sigma, SML1538) resuspended in DMSO for 6 hours. For transcriptional noise experiments, K562 cells were treated with 10uM 5-iodo-2′-deoxyuridine (IdU) (Sigma, 17125-5G) resuspended in DMSO for 24 hours and media containing 1mM 4sU (Sigma, T4509) was added during the last 8 minutes for subsequent nuclear isolation and nascent mRNA enrichment, as described below. Control cells were treated with equivalent amounts of DMSO without CHX, NMDi14, or IdU for 3, 6, and 24 hours, respectively.

METHOD DETAILS

Nuclear isolation

Nuclear isolation from total cells was performed following a previously established protocol.[74] Briefly, trypsinized cells were pelleted, washed with ice cold 1X PBS, and resuspended in ice cold Nuclear Isolation Buffer (10mM Tris-HCl, pH 7.4, 10mM NaCl, 3mM $MgCl_2$, 0.3% NP40, Roche EDTA-free protease inhibitor (Millipore Sigma, 11836153001), Sigma PhosSTOP phosphatase inhibitor (Millipore Sigma, 4906845001). The suspension was briefly centrifuged again to pellet the nuclei. The supernatant was removed (cytoplasmic fraction) and the pellet was resuspended in ice cold Nuclear Isolation Buffer. The suspension was centrifuged at 10,000 RCF and supernatant removed. The nuclear pellet was resuspended in TRIzol (Invitrogen, 15596018) before RNA isolation.

The purity of the nuclear isolation was confirmed using a Western blot. To do so, cells were washed with ice-cold PBS and resuspended in ice-cold RIPA buffer containing EDTA-free protease cocktail inhibitor. Cell membranes were ruptured by sonication and clarified by centrifugation at 14,000 x g for 15 mins at 4°C. 10–20 μg cell lysate was loaded in each well and resolved on a 4%–12% Tris-Glycine SDS PAGE and transferred to a nitrocellulose membrane. Briefly, the membrane was blocked in 5% milk, probed by antibodies (Cell Signaling Technologies, CST) 1:1,000 anti-vinculin and 1:1,000 anti-β-actin, followed by 1:2,000 anti-rabbit secondary (CST). HRP-linked secondary antibodies were visualized using ECL substrate (ThermoFisher).

RNA isolation and library preparation

For nascent RNA isolation from nuclei, K562 and KNS60 cells were incubated with 500uM or 1mM 4-thiouridine (Sigma, T4509), AM-38, HepG2 and SY5Y cells were incubated in 1mM 4-thiouridine, dissolved in water, for 8 minutes and spun down to pellet cells. The pellet was resuspended in Trizol and flash frozen in liquid nitrogen. After nuclear isolation and RNA isolation and described below, 4sU pulldowns were performed using MTSEA biotin-XX (VWR, 89139–636) 4sU-containing (nascent) transcripts were purified following previously established protocols.[75,76] Control cells were treated 1mM uridine (ChemGenes, RP-1186) dissolved in water for 8 minutes or with 1mM 4sU for 24 hours.

RNA was extracted from cells or nuclear pellets using TRIzol reagent to separate the aqueous RNA phase from the DNA and protein phases. The aqueous RNA phase was carried forward into column purification using a Zymo-25 RNA Clean & Concentrator kit (Zymo, R1018). Library preparation for mature RNA, CHX-treated, NMDi14-treated, and DMSO-treated samples was performed using the KAPA Stranded mRNA-Seq Kit for Illumina Platforms (KK8421). Library preparation for the nuclear and 4sU-labeled samples was performed using Kapa Biosystems KAPA RNA HyperPrep Kit with RiboErase (KK8561). Libraries were sequenced commercially at Novogene using an Illumina Novaseq or on an Illumina Nextseq 2000.

Synthetic spike-in libraries

Synthetic Spike-In RNA Variant (SIRV-Set1; Lexogen 025.03 & SIRV-Set4: Lexogen 141.01) was used to make libraries using both the stranded mRNA-seq and RNA HyperPrep kits, which were sequenced on NextSeq500. Resulting data was pre-processed, mapped, and analyzed as described below, using the SIRV reference annotations.

Cryptic splice site validation

Targeted amplicon sequencing was used to validate cryptic splice junctions identified from RNA-seq analyses. Primer pairs were designed with Primer3[77,78] to flank each candidate junction. K562 cells were grown in RPMI supplemented with 265 Units/mL penicillin, 265 μg/mL streptomycin and 10% FBS. RNA extracted from either whole cells or nuclear pellets was reverse-transcribed into cDNA using the High-Capacity cDNA Reverse Transcription Kit (Thermo Fisher Scientific, 4368813). PCR reactions were carried out with Phire Hot Start II PCR Master Mix (Thermo Fisher Scientific, F126) using 35 amplification cycles with an annealing temperature of 56 °C and a 15 s extension step. Amplified products were assessed on 2% agarose gels. Almost all the cryptic sites identified in nuclear RNA result in a correctly sized PCR product, of which 54% of low-fidelity and 75% of high-fidelity amplicons result in single-band products. While almost all high-fidelity sites identified in mature RNA result in a correctly sized PCR product (61% with a single band), only 21% of the low-fidelity sites are present as amplicons (75% with single band), likely due to their relatively low expression in mature RNA.

Selected amplicons with single bands were submitted to Genewiz (Azenta Life Sciences) for Amplicon-EZ next-generation sequencing and sequenced on an Illumina MiSeq platform using a 500-cycle paired-end configuration. Sequencing reads were aligned to the hg38 reference genome with ENSEMBL GRCh38.95 annotations using STAR[72] with parameters described previously. Cryptic splice junction usage was evaluated by identifying junction reads in the target regions.

Quantitative RT-PCR was used to quantify cryptic site usage. First, cDNA was synthesized from nuclear or total RNA using random hexamers or oligo(dT) primers, respectively. qPCR was performed using Luna Universal qPCR Master Mix (NEB, #M3003). Each 20 uL reaction contained 10 uL Luna Master Mix, 0.5uM of each primer and 0.5uL cDNA template. Thermal cycling condition was: initial denaturation at 95°C for 1 min, followed by 45 cycles of 95°C for 15 seconds and 60°C for 30 seconds. A melt curve analysis was performed at the end of amplification to confirm specificity. All reactions were performed with four technical replicates.

QUANTIFICATION AND STATISTICAL ANALYSIS

Pre-processing and quantification of reads

Adapters were trimmed and removed from reads with trimmomatic v0.32.[67] Adapter removal was verified by running fastqc v0.11.5[79] on trimmed fastqc files. Overlapping read-mates were merged with PEAR v0.9.11[68] and merged overlapping read-mates, non-overlapping read 1 and non-overlapping read 2 for each sample were pooled into a single fastq file per library. For all subsequent analyses, fastq files for cell lines were subsampled in two ways using seqtk v1.3[80] and then processed as described above: (1) (subsampling to 112.67M reads to match the depth of the library with the lowest sequencing depth and (2) subsampling to percentiles of the total number of reads 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80% and 90%) to analyze the sensitivity of cryptic site detection. Fastq files for K562 transcriptional noise experiments were subsampled to 33.1M reads to match the depth of library with the lowest sequencing depth. Fastq files for SIRV synthetic RNA were not subsampled. Transcripts per million (TPM) values were calculated using kallisto v0.45.0[70] (with parameters: –single—rf-stranded) on PEAR-processed, subsampled fastq files. Gene counts were generated from transcript counts from kallisto by tximport using EnsDB.Hsapiens.v86 annotations. Subsampled reads were subsequently processed using the *CRYPTID-SS* pipeline as described below. The distribution of intronic reads was estimated using bedtools coverage v2.28.0[71] (with -s parameter) for all introns expressed genes (TPM >= 2).

Cryptic sites identification using *CRYPTID-SS*

We developed a python-based framework to confidently identify cryptic splice sites from RNA-seq exon-exon junction reads. Briefly, this pipeline does the following five steps, each of which are detailed below as applied to the datasets analyzed in this manuscript: (1) modified read names in fastq files to facilitate downstream analyses, (2) map reads with STAR splicing-aware mapper to identify and quantify junction reads, (3) resolve strand discrepancies for non-strand-specific junctions, (4) identify gene-specific and genomic features associated with each junction, (5) filter out junctions which are likely the result of biological or technical artefacts. *CRYPTID-SS* takes raw fastq files as an input and outputs a filtered table that contains junction, splice site, and genomic features.

Step 1: Modifying read names

By default, STAR trims the second part of read names using a space delimiter. As a result, the read orientation information (read 1 vs. read 2) is not carried forward to BAM files. To retain this information to aid in strand assignment (below), the space character in read names was replaced by an underscore character in FASTQ files pre-mapping.

Step 2: Mapping reads

Reads with modified read names were mapped to the GRCH38 genome using STAR v2.7.0e[72] in single-end mode using default parameters with the following modifications or additions: *–outSAMattributes All—outSJfilterOverhangMin 40 12 12 12—outSJfilterCountUniqueMin 1 1 1 1—alignIntronMin 100—alignIntronMax 250000—limitOutSJcollapsed 2000000—outSJfilterDistToOtherSJmin 20 0 10 10—alignSJstitchMismatchNmax 2 2 2 2*. Reads are mapped in single-end mode since PEAR processing removes pairs for most reads. For all downstream steps, the resulting STAR junction database (.sj.out.tab) file was used.

Step 3: Resolve strand discrepancies

The STAR junction database does not contain strand information for junctions derived from sites with non-canonical unannotated intron motifs. To assign strand for these splice junctions, uniquely mapping junction reads were extracted for each site with a

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non-canonical intron motif. For each site, the number of uniquely mapped reads mapping to plus and minus strands were counted and the strand with the greater number of reads was assigned as the sense strand for that junction site.

Step 4: Identify gene-specific and genomic features associated with junctions

For each splice site within a junction, the following information was obtained by comparing to ENSEMBL GRCH38.95 annotations, which were parsed with *gtf2csv*[69]:

- Presence in annotations
- Location within a gene; ENSEMBL ID and strand if within a gene
- If within a gene, location within an annotated intron or exon including alternative or constitutive status of feature
- If within an exon, location within annotated UTRs, coding regions (CDS), or terminal codon
- Distance of splice site to nearest upstream 5'ss, nearest upstream 3'ss, nearest downstream 5'ss and nearest downstream 3'ss

To obtain sequence information, +/-25 nt of DNA sequence around the splice site and the relevant dinucleotide sequences (first 2nt downstream of 5' sites or first 2nt upstream of 3' sites) were extracted from the reference genome using *bedtools getfasta*. To estimate splice site strength, maxEnt scores for 5'ss (9nt region) and 3'ss (23nt region) were calculated using maxEntScan perl scripts that are modified to prevent skipping of sequences with "Ns" in them.[81] To estimate conservation, base-specific and mean phyloP scores were obtained for a +/- 20nt region around the splice site by applying the *bigWigToBedGraph* tool from *ucsctools* using the hg38.phyloP100way bigwig file downloaded from the UCSC database.[82,83]

Step 5: Filtering our spurious junctions

The following criteria were applied to remove spruious splice junctions before downstream analyses:

- Splice junctions with sites within annotated intergenic regions.
- Splice junctions whose 5′ and 3′ splice sites mapped to different genes.
- Splice junctions with 5′ and 3′ splice sites mapping to a distance of less than 25 nucleotides.
- Splice junctions with no uniquely mapped reads.
- Splice junctions whose splice sites map within and near (within 10nt) the same repeat regions.
- Splice junctions whose 5′ and 3′ splice sites map within the same type of repeat within the same gene.
- Splice junctions containing polymorphisms in the first two or last two nucleotides of the intron (see genotyping section below).
- For K562 datasets, splice junctions containing individual splice sites that were also identified in the K562 exome (ERR2990087)[66] dataset were removed.

Identifying repeat-associated splice junctions. Bed file containing DNA repetitive elements for GRCh38 was downloaded from UCSC table browser. Splice junctions completely within repeat regions were identified with GenomicRanges using *findOverlaps* (*type* = "*within*"). Splice junctions near repeats with identified with GenomicRanges *findOverlaps*(*type* = "*within*", *maxgap=10*).

Genotyping of cell lines from RNA-seq data. To ensure compatibility with GATK tools,[73] MAPQ scores in BAM files generated from STAR were changed to "60" from "255", and readgroups were added by *AddOrReplaceReadGroups* (Picard v2.27.5).[84] Junction reads were split using *SplitNCigarReads* (GATK v4.3.0.0).[85] Haplotypes were called by *HaplotypeCaller* with the flag "*-ERC GVCF*" (GATK v4.3.0.0) and GVCF files from nuclear, nascent and polyA samples for each cell type (HepG2, K562, KNS60, AM38 and SY5Y) were combined using *CombineGVCFs* (GATK v4.3.0.0). Variants were cataloged using *GenotypeGVCFs* (GATK v4.3.0.0). Finally, variant .vcf files were converted to table format with *VariantstoTable* (GATK v4.3.0.0).

Defining cryptic splice sites. Splicing junctions identified by *CRYPTID-SS* were collapsed into a unique set of splice sites, retaining junction pairing information for downstream analyses and position within the junction (5′ or 3′ site). Cryptic sites were categorized as belonging to one of three subtypes. High-fidelity cryptic junctions were defined as splice sites with canonical dinucleotides but have not been previously annotated as splice sites. These sites are derived from reads with SAM flag attributes of $jM:B:c,[1,2,3,4,5,6]$. Recursive sites were defined as those falling within an "AGGT" sequence motif necessary for recursive splicing, with 5′ recursive sites defining an intron beginning with the "GT" and 3′ recursive sites defining an intron ending with the "AG". These sites are derived from reads with SAM flag attributes of $jM:B:c,[1,2]$ and additionally have a templated 3′ or 5′ dinucleotide upstream of downstream of the 5′ or 3′ site, respectively. Low-fidelity cryptic sites were defined as splice sites without canonical dinucleotides (*i.e.* the first two nucleotides of the intron are not "GT" or "GC" or "AT" (for 5′ sites) and last two nucleotides of intron not containing "AG" or "AC" (for 3′ sites)). These sites are derived from reads with SAM flag attributes of $jM:B:c,[0,20]$.

Public K562 DNA-sequencing datasets

K562 whole genome sequencing data associated with accession number ENCSR045NDZ[53] were downloaded. Samples were mapped and processed as described above to identify "junction" reads.

Simulated pre-mRNA and mRNA reads

Synthetic reads derived from pre-mRNA (unspliced) and mRNA (spliced) transcripts from 5000 genes were generated using SPARK.[86] We simulated two replicates per condition, using a fixed expression level of 150 TPM per gene, mean library insert size of 268 nt, and 150nt paired end reads. Simulated reads were mapped as described above and *CRYPTID-SS* was used to identify cryptic sites. Simulation data was used to estimate false positive rates (FPR) for errors introduced during library preparation and sequencing. Specifically, the FPR was calculated as the proportion of the false splicing instances (number of junction reads mapping to cryptic sites) among all non-junction reads.

Comparing mapping accuracy with BLAT

Reads containing canonical, high-fidelity novel and low-fidelity novel splice junctions were isolated using SAM flag attributes of $jM:B:c,[21]$, $jM:B:c,[1,2,3,4,5,6]$, and $jM:B:c,[0]$ respectively. BAM files containing these reads were converted to FASTQ files with samtools[87] *bam2fq* v1.9, which were then converted to FASTA files. Reads in FASTA files were mapped with BLAT[88] v37x1 with these parameters: *-q=rna -fine -noHead*. For comparison with BLAT mapped reads, the same reads from STAR were converted to bed files with Bedtools *bam2bed* v2.28.0 with with this parameter: *-cigar*.

Modeling usage of individual splice sites

To estimate the extent to which genomic factors at or around each splice site accounted for variance in the usage of splice sites, we fit a linear model of the following form, separately for canonical, low-fidelity, recursive, and high-fidelity splice sites:

$$y_i = \beta_0 + \beta_{i1}x_{i1} + \beta_{i2}x_{i2} + \beta_{i3}x_{i3} + \beta_{i4}x_{i4} + \beta_{i5}x_{i5} + \beta_{i6}x_{i6} + \varepsilon_i$$

where y_i reflects the usage of a splice site (log10, reads), x_{i1} is conservation score (mean phyloP score in a 50nt region around the splice site), x_{i2} is splice site score (from maxEnt), x_{i3} is the length of the intron or exon within which the site is in (log10, nucleotides), x_{i4} is the number of partner sites detected, x_{i5} is the mean distance between splice site partners (log10, nucleotides), and x_{i6} is the fractional position of the sites within the parent gene. We used the values from this multiple linear regression model to estimate the relative importance of each parameter contributing to variance in splice site usage. To do so, we used the *relaimpo* package in the R statistical environment,[89] which arrives at a relative importance percentage by averaging the sequencing sum-of-squares obtained from all possible orderings of the predictors in the model.

Co-transcriptional splicing analysis

The extent of co-transcriptional splicing that occurs at cryptic splice sites was estimated with individual indices for introns and exons. The completed splicing index (coSI)[90] was calculated with read counts that span across exon boundaries into the adjacent intron (incomplete splicing) relative to read counts of split reads across exon-exon junctions (complete splicing). The coSI was used to assess co-transcriptional splicing when cryptic sites occurred in an intronic region. The splicing per intron (SPI)[91] was calculated with junction read counts for spanning an exon-exon junction (completed splicing) relative to the total number of reads spanning exon-exon and exon-intron junctions (incompleted splicing). The SPI was used to assess co-transcriptional splicing when cryptic sites occurred in an exonic region.

Sequence content analyses

maxEnt scores

Splice site strengths were estimated using a maximum entropy model as implemented in maxEntScan[81] using 9 bp around the 5′ splice site (−3: +6) and 23 bp around the 3′ splice site (−20:+3).

PWM scores

To score splice site strength excluding core dinucleotides, we first derived 5′ and 3′ splice site position weight matrices by running MEME[92] on the [-6:+3] and [-20:+3] regions around canonical 5′ and 3′ splice sites from 10,000 randomly sampled human introns (as described in [93]). PWM scores (PWMS) were then computed as described in [42], considering each position to be independent and against a background matrix with equal probability of each nucleotide at every position.

SRE enrichments

To assess the enrichment of known splicing regulatory elements (ESE, ESS, ISE, ISS),[94–97] we extracted 50-nt sequence windows upstream and downstream of splice sites and calculated hexamer frequencies. Hexamer frequencies from randomly sampled 50-nt regions from annotated transcripts (GENCODE v49) were used as a background set and significantly enriched kmers were identified using Fisher's exact test followed by a Benjamini–Hochberg FDR correction. The enrichment of each SRE class was quantified as the ratio of significant hexamer fractions in each SRE category relative to non-SREs and significant enrichment was evaluated using a Hypergeometric Test.

Dinucleotide analyses

All donor and acceptor dinucleotide pairs were extracted from the reference genome (either human or the SIRV reference sequence) and the expected distribution of sites was calculated as the frequency each dinucleotide occurs in expressed genes. To estimate significant enrichments of observed dinucleotide pairs, observed dinucleotide pairs were selected at random from the reference genome and then randomly shuffled. This shuffling process was repeated 1000 times to calculate a probability for each possible

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permutation in order to generate a null distribution. This null distribution was then used to calculate the statistical probability for the occurrence of dinucleotide pairs and identify significantly enriched dinucleotide pairings in the observed data.

Quantifying cryptic splicing transcripts

To estimate how many transcripts for each gene had type of cryptic splicing, we calculated TPMs using only junction reads, separated by cryptic type. We assumed that each transcript has at most one cryptic splicing event, and thus cryptic junction reads were treated as an individual transcript. Enrichment method-specific RPKMs were then calculated for each splice site category individually, where R is junction reads, K is the effective transcript length in kilobases estimated by *kallisto*, and M is the total number of junction reads (in millions). RPKMs for each gene were then summed up per sample to calculate junction-specific transcripts per million ($TPM_{junction}$) as follows: $TPM_{junction} = 10^6 * \frac{RPKM}{\Sigma RPKM}$.

Elastic net regression modeling for genes

To build a predictive model for the extent of cryptic splicing per gene, we trained an elastic net regression model using the *caret* package in the R statistical environment.[98] Briefly, we used the *glmnet* method to tune both the λ and α parameters using five-fold cross-validation and data were centered and scaled prior to use in the modeling. The variables modeled were transcription levels (log10, TPMs from nascent RNA), steady-state gene expression levels (log10, TPMs from mature RNA), gene length (log10, nucleotides), the number of introns in the gene, and the percentage of a gene that is intronic. Elastic net regression was chosen to select predictive features and avoid overfitting with variables that are highly correlated to each other. Expected values for each gene were predicted from the final model using the *predict* function in the R statistical environment and R^2 values were calculated as the Pearson correlation between the predicted and observed values for each gene. Models were fit separately to data from each enrichment method and each category of splice sites (canonical, low-fidelity, recursive, and high-fidelity sites). To identify genes with more cryptic splicing than expected given the variables modeled, studentized residuals were calculated from the model residuals and any gene with a studentized residual greater than 1.5 was categorized as having more cryptic splicing than expected for that cryptic splicing category.

Gene Ontology

Gene ontology analysis was performed using WEB-based GEne SeT AnaLysis Toolkit[99] to perform gene set enrichment analysis for genes in the categories defined by junction TPMs and predicted outliers characterized by the elastic net regression model.

Analysis of publicly available splicing inhibition data

FASTQ files of RNA-seq and TT-seq samples from either 1uM or 100nM of Pladienolide B (PlaB) treatment, respectively, of K562 cells were downloaded from GSE148433[51] along with matched DMSO controls. Reads were trimmed and processed with PEAR. RNA-seq samples were subsampled to 28M reads, and TT-seq samples were subsampled to 75.8M reads. Subsampled reads were mapped with STAR and cryptic splice sites were identified with *CRYPTID-SS*, both as described above.

Analysis of publicly available knockdown data

FASTQ files for shRNA knockdown RNA-seq samples against 304 splicing related (SF) proteins in HeLa cells[52] were downloaded from ArrayExpress (E-MTAB-11202) and FASTQ files for shRNA knockdown RNA-seq samples against RBPs in K562 cells were downloaded from the ENCODE portal.[53] FASTQ files from read1 and read2 were processed with PEAR, subsampled to 131.8M or 12.7M reads (for SF KD or RBP KD respectively), and mapped with STAR as described above. Cryptic splice sites were identified with *CRYPTID-SS* and filtered as described above. RNA-seq libraries from SF KDs were un-stranded, so low-fidelity sites were assigned to the strand of their parent gene after relevant filtering steps. To determine statistical significance, control shRNA samples were compared to other randomly selected control shRNA samples 100 times to produce null distributions of the proportions of cryptic sites and cryptic junction reads. The observed proportion of cryptic sites or reads in the KD samples were compared to the null distribution to determine the probability of observing these proportions by chance. FDRs for p-values were calculated using the Benjamini Hochberg correction. For splicing related protein KD samples, the proportions of cryptic splice sites and cryptic junction reads in each knockdown were compared with mean values across the seven control samples. The permuted null distribution was created using all 27 pairwise comparisons of the seven control samples, resulting in a minimum achievable p-value of 0.04. For RBP-KD samples, the proportions of cryptic splice sites and cryptic junction reads in each knockdown were compared with the batch-matched scrambled shRNA control sample. The permuted null distribution was created using 100 random pairwise combinations of the 52 control samples, resulting in a minimum achievable p-value of 0.01.

Analysis of RNA flow data

To assess the lifecycle of genes with more cryptic sites than expected, we used subcellular residence times and half-lives estimated for each gene in K562 cells from the RNAflow method as found in Table S1 from Ietswaart et al.[57] predicted to undergo nuclear degradation (PUND) classification for each gene was also obtained from Ietswaart et al.[57]

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Supplemental information

Pervasive noise in human pre-mRNA splice site selection

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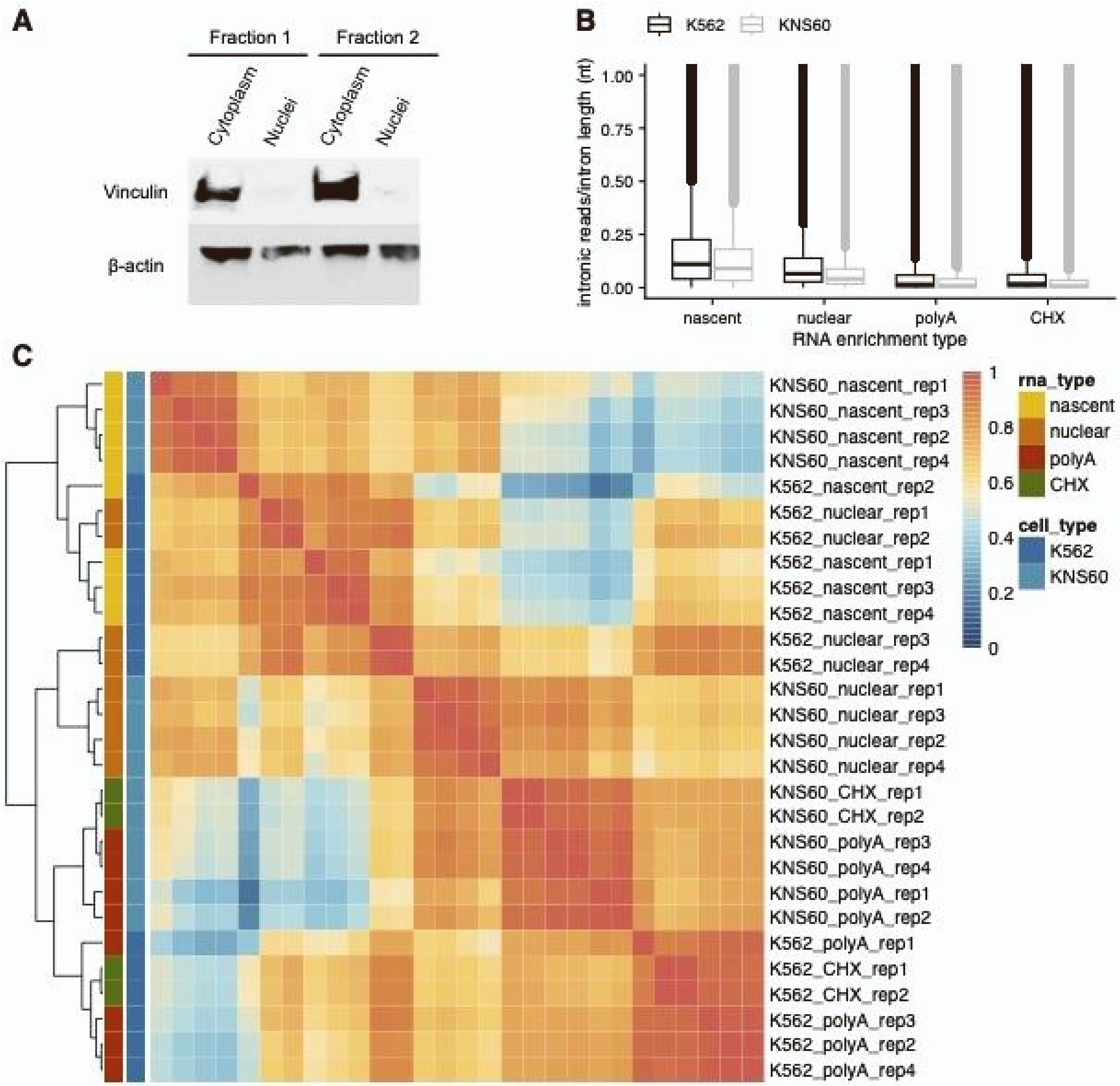


Figure S1: Validation of RNA enrichment methods, related to Figure 1. (A) Western blot of fractionated cell samples. Vinculin, a membrane-cytoskeletal adhesion protein, is absent in purified nuclear fractions but present in cytoplasmic fractions. β-actin was used as a loading control. **(B)** Distribution of intronic reads (normalized to intron length, *y-axis)* across RNA enrichment methods *(x-axis)* for K562 *(black)* and KNS60 *(gray)* cells. **(C)** Clustering heatmap of TPM values for each sample across cell types and RNA enrichment methods, with a hierarchical clustering dendrogram shown on the left.

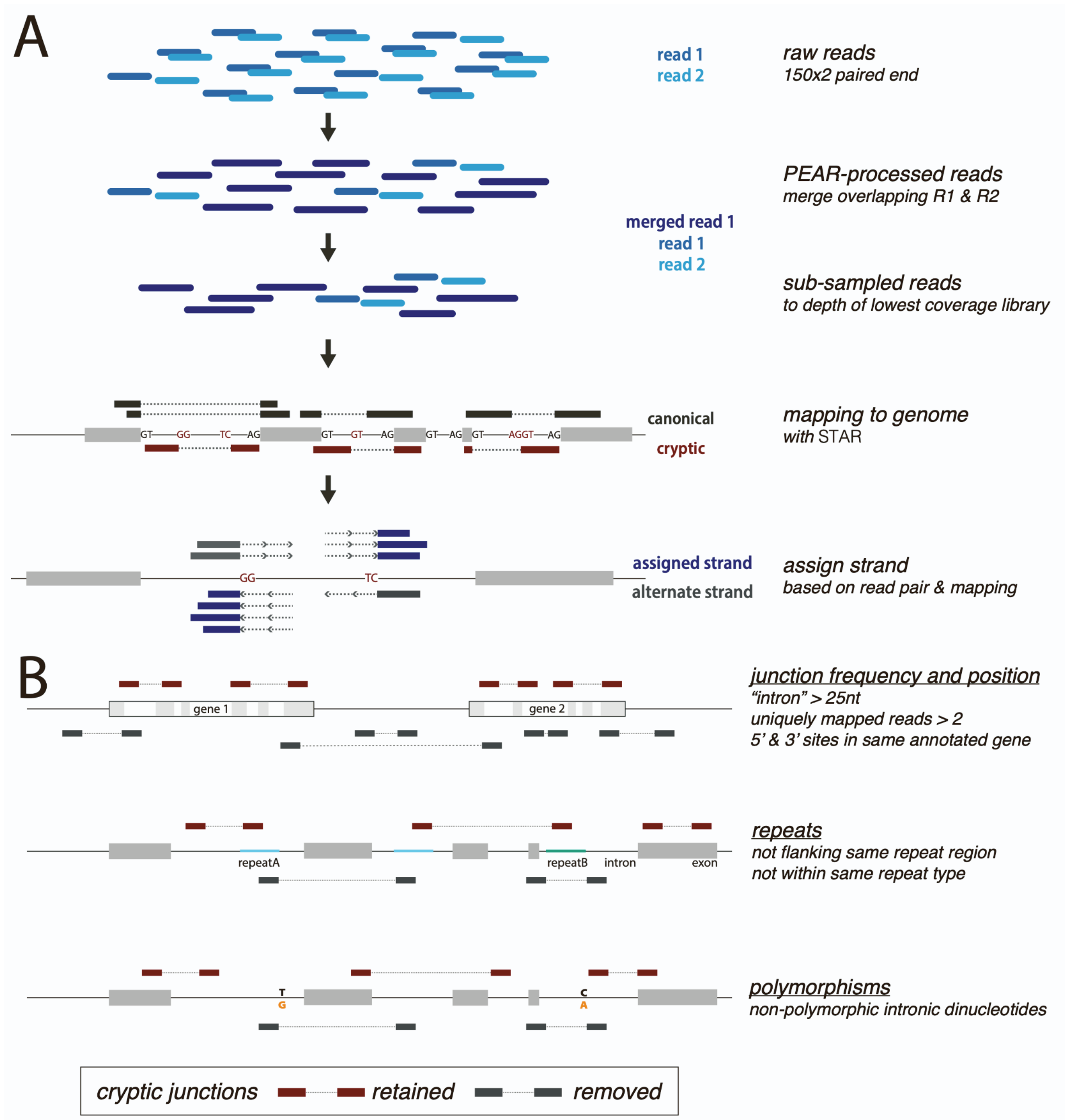


Figure S2: Computational pipelines and criteria used to identify cryptic splice sites, related to Figure 1. (A) Schematic representation of the computational pipeline to identify cryptic splice sites, including collapsing overlapping read pairs, subsampling read depth, mapping and annotation, and strand assignment for novel sites. **(B)** Schematic representation of the criteria necessary to identify cryptic splice sites, including junction frequency and position, proximity to repeat region(s), and overlap with sequence variation.

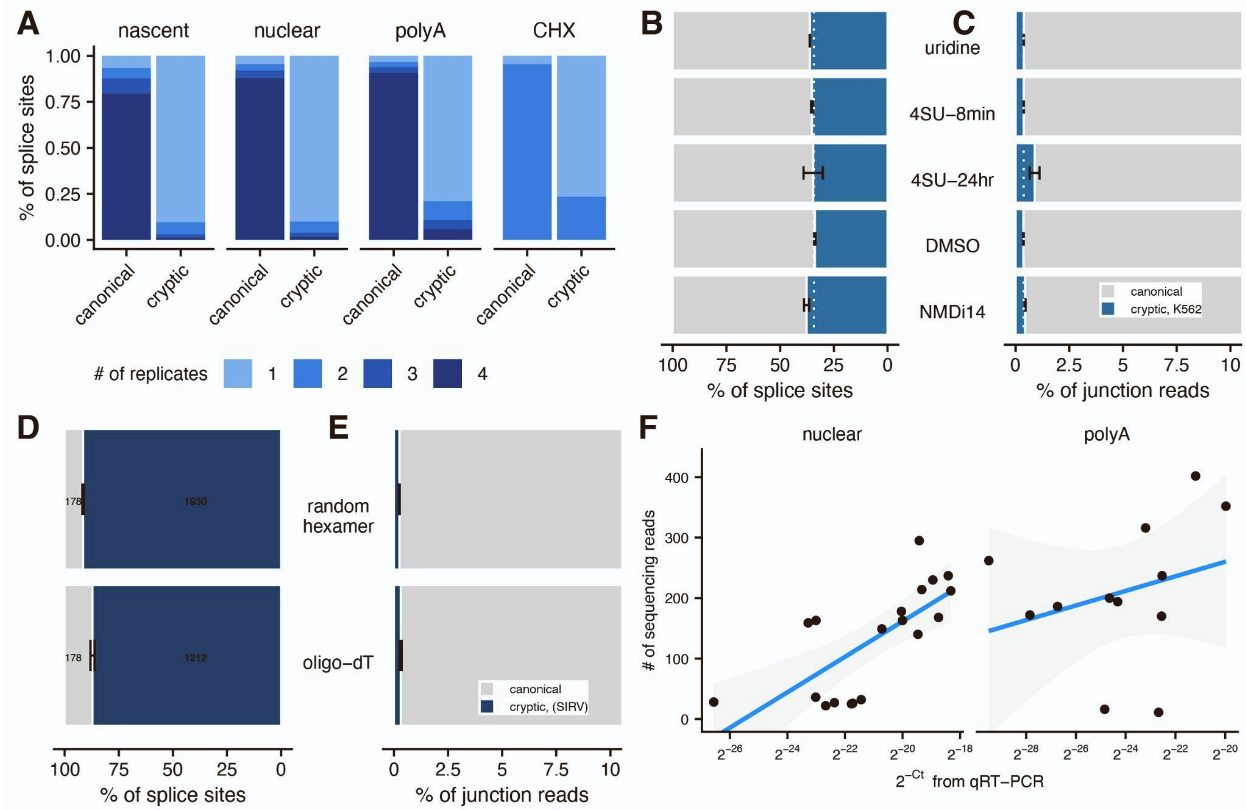


Figure S3: Control datasets used to establish confidence in cryptic site identification and quantification, related to Figure 1. (A) Percentage of canonical or cryptic splice sites that were detected in 1, 2, 3, or all 4 replicates across enrichment methods. Percentage of canonical (*grey*) and cryptic (*blue*) splice sites **(B)** and junction reads **(C)** detected across control treatments in K562 cells. Dotted white line represents the mean percentage for polyA-enriched samples. Percentage of canonical (*grey*) and cryptic (*blue*) splice sites **(D)** and junction reads **(E)** detected in random hexamer or oligo-dT primed SIRV libraries. (F) The relative expression of cryptic sites estimated from qRT-PCR (*x-axis*) versus high-throughput sequencing (*y-axis*) for sites identified in nuclear (*left*) or mature RNA (*right*).

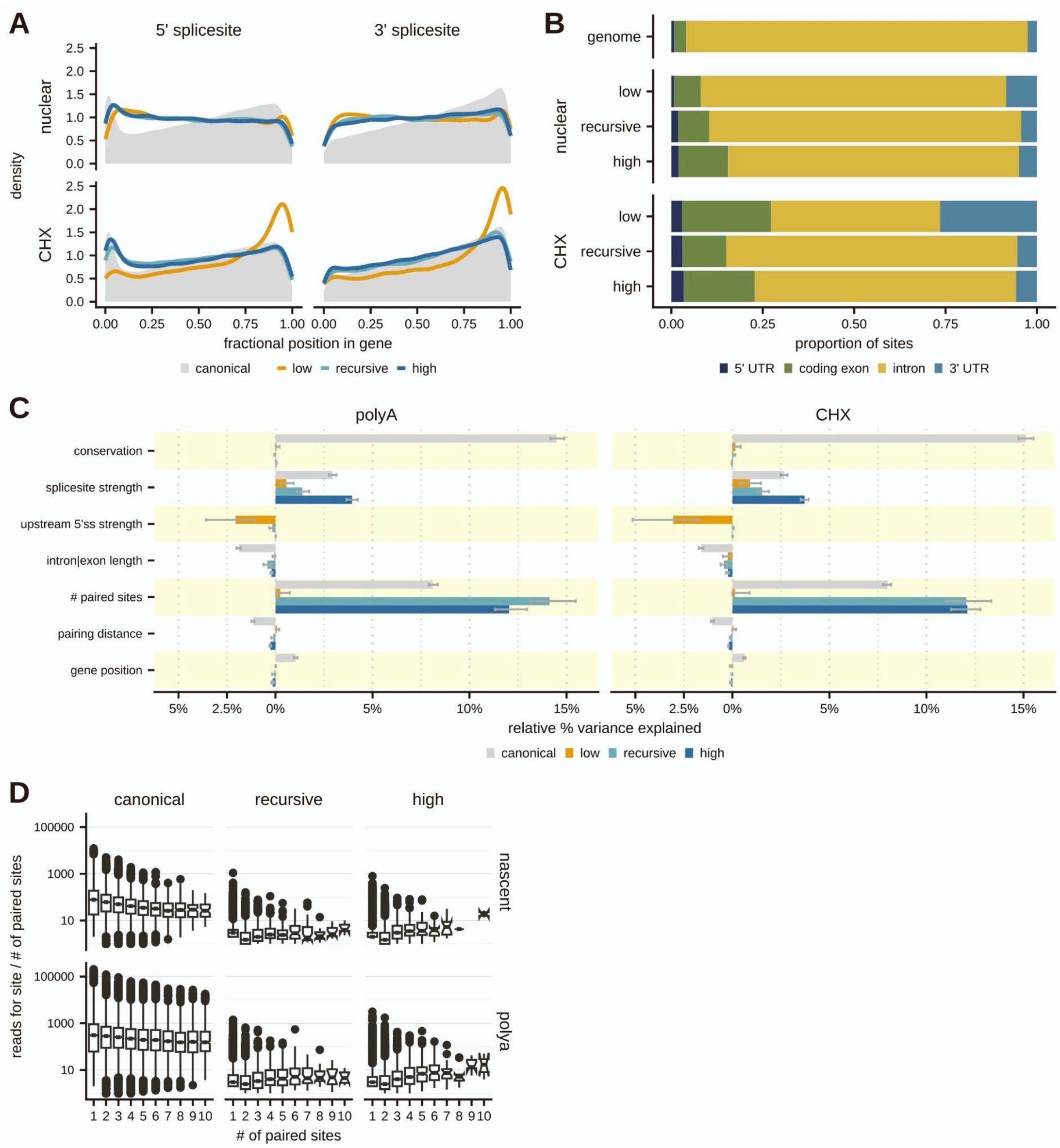
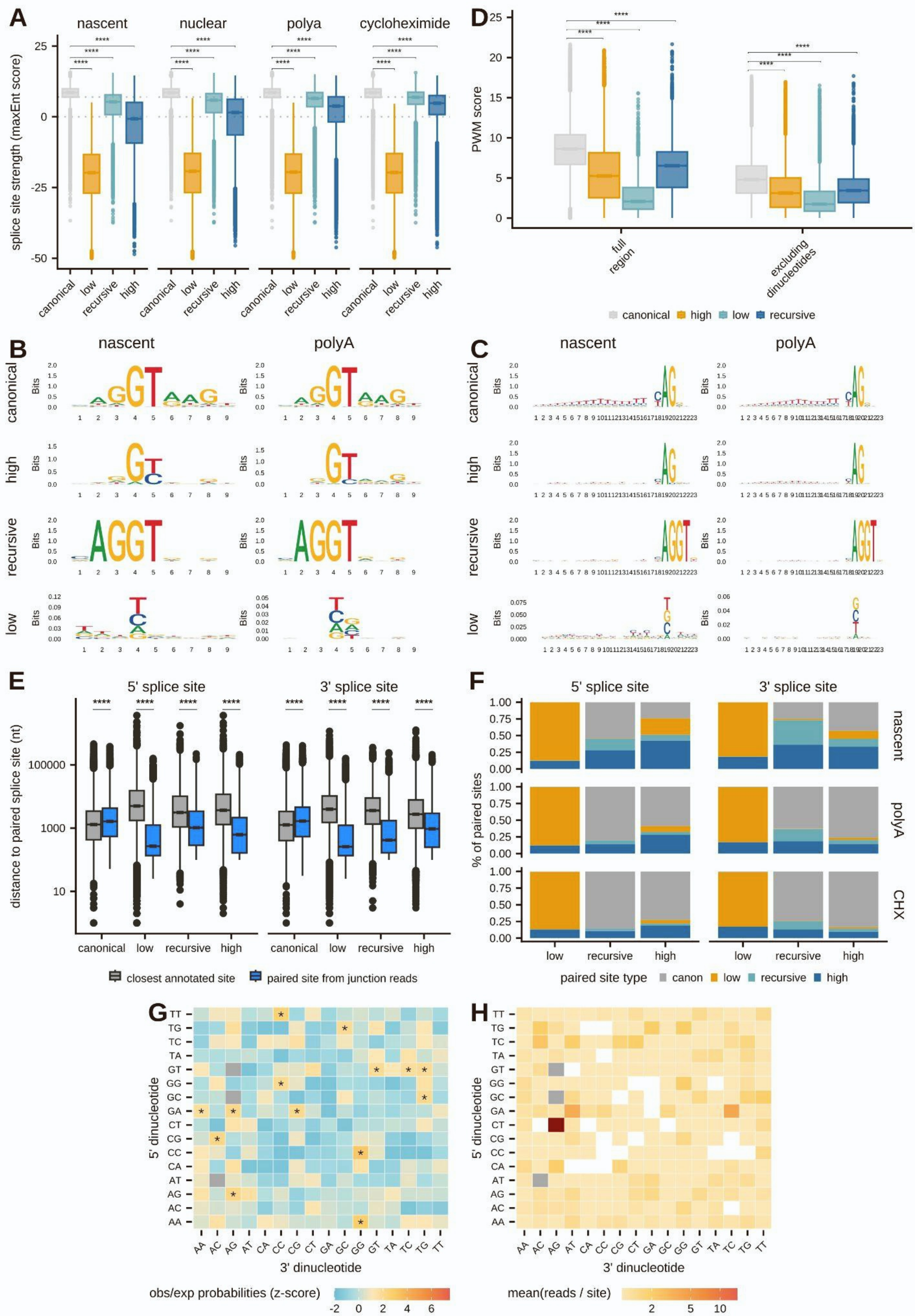
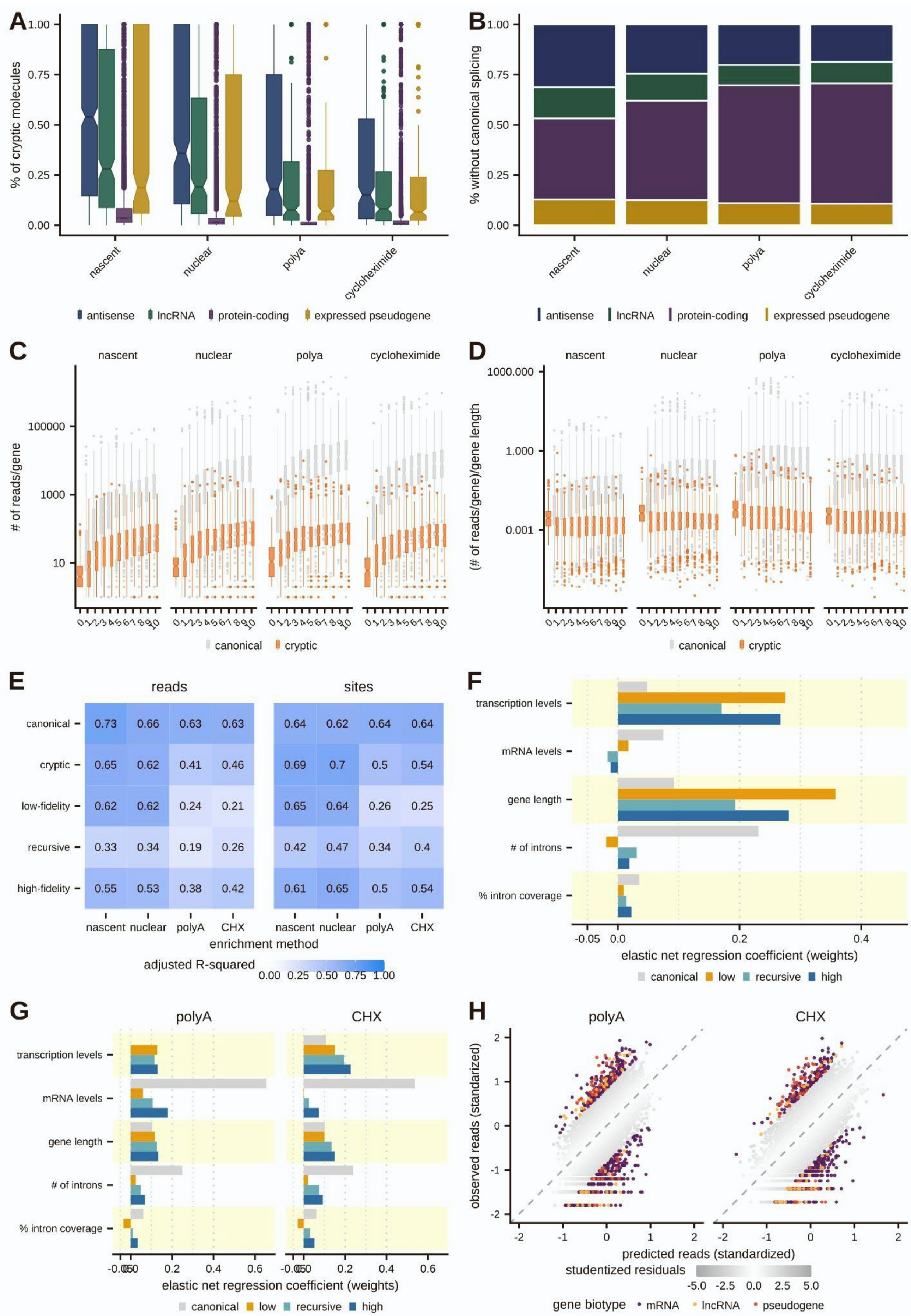


Figure S4: Genomic features of cryptic sites across RNA enrichment methods, related to Figure 2. **(A)** Distribution of the position of 5' (*left*) and 3' (*right*) splice sites across a gene categorized as canonical (*grey*), low-fidelity (*orange*), recursive *(teal)*, or high-fidelity (*dark blue)* in either nuclear (*top*) or CHX-treated (*bottom*) RNA, where the fractional position is defined as the distance of a site from the start of a gene relative to the total gene length. **(B)** The proportion of each type of cryptic site detected within annotated 5' UTRs (*navy blue*), coding exons (*green*), introns (*yellow*), and 3' UTRs (*light blue*) for sites detected in nuclear (*middle*) or CHX-treated (*bottom*) RNA. The top bar shows the proportion of nucleotides within expressed genes for each annotation category. **(C)** Relative importance of features influencing variance in splice site usage for canonical (*grey*), low-fidelity (*orange*), recursive (*teal*), and high-fidelity (*dark blue*) sites, using multiple linear regression models for mature *(left)* and CHX-treated RNA *(right)*. Bars pointing to the right indicate positively correlated features, while bars pointing left indicate negatively correlated features. **(D)** Distribution of the number of reads relative to the number of paired sites for the same site *(y-axis)* across splice sites with increasing numbers of paired sites *(x-axis)* for canonical *(left)*, recursive *(middle)*, and high-fidelity *(right)* splice sites in nascent *(top)* and mature *(bottom)* RNA. Very few low-fidelity sites have more than one pairing partner.



A
nascent
nuclear
polya
cycloheximide
splice site strength (maxEnt score)
canonical
low
recursive
high
D
PWM score
full region
excluding dinucleotides
canonical
high
low
recursive
B
nascent
polyA
canonical
high
recursive
low
Bits
C
nascent
polyA
E
5' splice site
3' splice site
distance to paired splice site (nt)
closest annotated site
paired site from junction reads
F
5' splice site
3' splice site
% of paired sites
nascent
polyA
CHX
paired site type
canon
low
recursive
high
G
5' dinucleotide
3' dinucleotide
obs/exp probabilities (z-score)
H
5' dinucleotide
3' dinucleotide
mean(reads / site)

Figure S5: Pairing characteristics of canonical and cryptic splice sites, related to Figure 3. (A) The distribution of splice site strengths (maxEnt scores, *y-axis*) for canonical (*grey*), low-fidelity (*orange*), recursive (*teal*), or high-fidelity (*dark blue*) splice sites across RNA enrichments. Significance was assessed with a Mann-Whitney U Test. **** adjusted p-value < 0.0001. **(B)** Seqlogos for canonical, high-fidelity, recursive, and low-fidelity 5' splice sites identified in nascent (*left*) and mature RNA (*right*). **(C)** Seqlogos for canonical, high-fidelity, recursive, and low-fidelity 3' splice sites identified in nascent (*left*) and mature RNA (*right*). **(D)** The distribution of PWM scores including (*left*) and excluding (*right*) the core dinucleotides for canonical (*grey*), low-fidelity (*orange*), recursive (*teal*), or high-fidelity (*dark blue*) splice sites in nascent RNA. Significance was assessed with a Mann-Whitney U Test. **** adjusted p-value < 0.0001. **(E)** The distribution of distances between pairing partners (blue) or the closest annotated splice site (grey) for 5' *(left)* and 3' *(right)* splice sites divided by canonical, low-fidelity, recursive, and high-fidelity sites. Significance was assessed with a Mann-Whitney U Test. **** adjusted p-value < 0.0001 **(F)** Proportion of 5' *(left)* and 3' *(right)* splice sites divided by cryptic type (*x-axis*) that pair with canonical *(gray)*, low-fidelity *(orange)*, recursive *(teal)*, or high-fidelity *(blue)* site in nascent *(top)*, mature *(middle)*, and CHX-treated *(bottom)* RNA. **(G)** Heatmap of the ratio between the observed number of sites relative to the expected distribution of sites (calculated by estimating the frequency each dinucleotide occurs in the SIRV transcriptome) for each combination of 5' (*y-axis*) and 3' (*x-axis*) dinucleotides used by non-annotated splice sites detected in SIRV libraries. Dinucleotide pairs involved in canonical splice site pairings were omitted from the plot (*grey boxes*). Significance was assessed using permutations. * adjusted p-value < 0.05. **(H)** Heatmap of the mean number of reads per site for each combination of 5' (*y-axis*) and 3' (*x-axis*) dinucleotides used by non-annotated splice sites detected in SIRV libraries. Dinucleotide pairs involved in canonical splice site pairings and pairs with fewer than 25 observations were omitted from the plot (*grey* and *white boxes*, respectively).



A
B
% of cryptic molecules
% without canonical splicing
nascent
nuclear
polya
cycloheximide
antisense
lncRNA
protein-coding
expressed pseudogene
C
D
# of reads/gene
(# of reads/gene)/gene length
canonical
cryptic
E
reads
sites
canonical
cryptic
low-fidelity
recursive
high-fidelity
nascent nuclear polyA CHX
enrichment method
adjusted R-squared
F
transcription levels
mRNA levels
gene length
# of introns
% intron coverage
elastic net regression coefficient (weights)
canonical low recursive high
G
polyA
CHX
H
observed reads (standarized)
predicted reads (standardized)
studentized residuals
gene biotype
mRNA
lncRNA
pseudogene

Figure S6: Elastic net regression to predict cryptic splicing across RNA enrichment methods, related to Figure 4. (A) The distributions of the percentage of transcripts with cryptic splicing (*y-axis*) across different gene biotypes (*colors*) in RNA enrichments (*x-axis*). **(B)** The proportion of genes without any canonical splicing (*y-axis*) that fall into different gene biotypes (*colors*) across RNA enrichments (*x-axis*). **(C)** The distribution of canonical (*grey*) or cryptic (*orange*) junction reads per gene (*y-axis*) across genes with different numbers of introns (*x-axis*) in different RNA enrichments. **(D)** The distribution of canonical (*grey*) or cryptic (*orange*) junction reads per gene normalized by gene length (*y-axis*) across genes with different numbers of introns (*x-axis*) in different RNA enrichments. **(E)** The adjusted R-squared values for elastic net regression models trained on different splicesites (*y-axis*) in each enrichment method (*x-axis*) using the number of junction reads *(left)* or splice sites *(right).* **(F)** Feature weights from elastic net regression models (*x-axis*) trained on the number of cryptic splice sites per gene for canonical (*grey*), low-fidelity (*orange*), recursive (*teal*), and high-fidelity (*dark blue*) sites in nascent RNA. Features are described in Figure 4A. **(G)** Feature weights from elastic net regression models (*x-axis*) trained on the number of cryptic splicing reads per gene for canonical (*grey*), low-fidelity (*orange*), recursive (*teal*), and high-fidelity (*dark blue*) sites in polyA *(left)* and CHX-treated *(right)* RNA. Features are described in Figure 4A. **(H)** The amount of cryptic splicing per gene predicted by the elastic net regression model (center standardized, *x-axis*) versus the observed cryptic splicing per gene (*y-axis*) in polyA *(left)* and CHX-treated *(right)* RNA. The grey color scale indicates the magnitude of studentized residuals and genes with |studentized residual| ≥ 2 are colored by their ENSEMBL biotype.

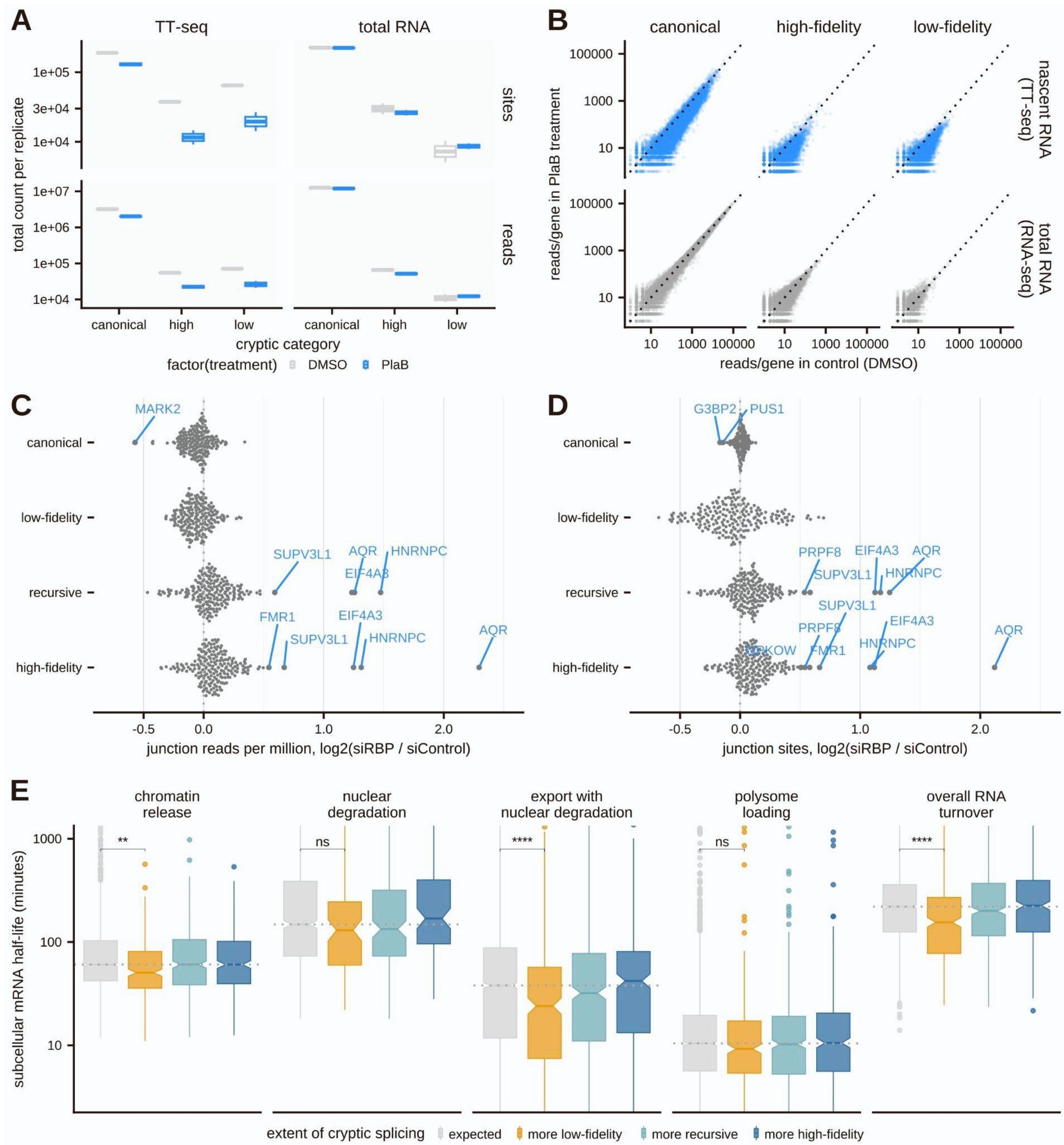


Figure S7: Identifying factors that influence cryptic splicing or the fate of cryptic transcripts, related to Figure 5. (A) The distributions of the total number (*y-axis*) of junction sites (*top*) or reads (*bottom*) in nascent RNA (TT-seq, *left*) or total RNA (RNA-seq, *right*) data across replicates following DMSO (*grey*) or PlaB (*blue*) treatment in K562 cells. **(B)** The correlation between the number of canonical, high-fidelity, or low-fidelity junction reads per gene in nascent RNA (TT-seq, *top*) or total RNA (RNA-seq, *bottom*) between DMSO (*x-axis*) and PlaB-treated (*y-axis*) cells. **(C)** The fold change in junction reads between RNA-binding protein siRNA knockdown and scrambled siRNA control samples (*x-axis*) for canonical, low-fidelity, recursive, and high-fidelity junction reads across all RBPs assessed by the ENCODE consortium. Significance was estimated using a permuted null distribution using control samples and significantly changing factors are labeled in blue (adjusted p-value < 0.05). **(D)** The fold change in junction sites between RNA-binding protein siRNA knockdown and scrambled siRNA control samples (*x-axis*) for canonical, low-fidelity, recursive, and high-fidelity junction reads across all RBPs assessed by the ENCODE consortium. Significance was estimated using a permuted null distribution using control samples and significantly changing factors are labeled in blue (adjusted p-value < 0.05). **(E)** Distributions of mRNA residence times or half-lives across additional cellular compartments or fractions estimated using subcellular TimeLapse-seq from Ietswaart *et al.* 2024 for genes with more low-fidelity (*orange*), recursive (*teal*), or high-fidelity (*dark blue*) transcripts than predicted by the elastic net regression model, with genes whose cryptic splicing levels match expected levels shown in grey. Significance was assessed with a Mann-Whitney U Test. **** adjusted p-value < 0.0001.